Honors Physics @ TJ

Abstract

These are notes for the Honors Physics course at TJHSST.

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1 Particle Kinematics

Unfortunately, I'm starting these class notes in Q2 instead of have them from the beginning, but better late than never, as they say... Hopefully, I will have time to come back to the kinematics we treated in so much detail in Q1, but in the meantime, here are the equations.

1.1 Space, time, and symmetry

Before we can talk about the laws of motion, we first need to talk about what motion is. In order to do that, we must first address what we can agree about the where and when of events. We'll start with the when.

Time When we talk about events, it is important to establish when they happened. Without thinking much about it, we casually agree on the "now". No units are needed for this.

To talk about times that are not now, we have argued for centuries about which standards we should use, eventually settling (at least for the purposes of science) on the unit of seconds.[1] Once this is done, we can talk about events that will happen "ten seconds from now" or "a minute ago".

A device that can measure changes in time is called a clock. Good clocks will "keep time" by always running at the same rate: A good clock does not speed up or slow down in the process of making measurements, but this is possible only if the progression of time is uniform and universal. We take this regular progression of time for granted, and it is *an assumption* in Newtonian physics that time is that way.

We also *assume* for the purposes of Newtonian physics, that time has always been the way it is now. In particular, there is no "origin of time" before which time did not exist. When we talk about $t = 0$, it is understood that we have agreed upon a convention for when to start the clock; there is no implication that times before that do not exist. (Although it can be that the physical situation under study underwent some substantive change when passing from earlier times to $t \geqq 0$. In this sense, there is usually also some final time after which something changes.)

In summary: Time t is not absolute, but changes in time Δt are measurable. A second system of time t' is considered equivalent to the first if they differ only in origin (*i.e.* what is being called 0 in either system). In other words, they will be equivalent if there is some real constant $c \in \boldsymbol{R}$ such that $t' = t + c$. When this happens, two events f and i will be separated in time by the same amount:

$$\Delta t' = t'_f - t'_i = (t_f + c) - (t_i + c) = t_f - t_i = \Delta t$$

[1] This is not as simple as it sounds. Historically, we took 1 second to mean $1/86400$ of a day, but this is way too inaccurate a measure. The modern definition involves a very specific transition in the electron orbitals of a cesium atom.

On a more mathematical note: We can identify the collection $\boldsymbol{T} = \{t\}$ of all valid times with the real numbers $\boldsymbol{R}$, but to do so, we have to pick which time will be identified with the number 0 and which with the number 1. After that, regularity in time will line up with the analogous regularity in the real number system. Picking these two things "selects an origin" and "sets the scale" we call seconds. Mathematically, we are saying that after you have picked these two pieces of information, you may identify $\boldsymbol{T} = \boldsymbol{R}$. [2]

Space As humans we take for granted that we can move in 3 independent directions we call "dimensions": forward/backward, left/right, and up/down. Each of these directions is just like the others (except that we find it much harder to move in the third one because of gravity, but we'll get to that).

Many of the physical phenomena we will study take place in only 2 or even 1 of the 3 dimensions available. For example, a ball falling from rest under the influence of gravity moves only in the vertical direction. Because of this and the previous remark, we focus first on the case where all of space is just one dimension.

One-dimensional (1D) space is very similar to time: Similarly to "now", we have "here", and similarly to "1 second from now" we have "1 meter from here": Again, we had to decide on a convention for how we will measure distance. Analogously to the clock, there is a device to measure distance called a "ruler".

Also, just as with time, there is no "origin of space" and any point is in this space is just like any other. This again makes it different from $\boldsymbol{R}$ which has a special point called $0 \in \boldsymbol{R}$. Just as with time, we can pick a point in space to identify with this special 0. We can then pick a point a distance of 1 meter from the first and send that to $1 \in \boldsymbol{R}$. This fixes where all the other points go, and we again have that 1D space can be identified with $\boldsymbol{R}$, but notice that this is an independent identification from the one we made for $\boldsymbol{T}$. (It may help to think of each as having its own copy of $\boldsymbol{R}$ for this.)

Finally, note that–just as with time–two coordinate systems for 1D space will be equivalent if they differ only by a translation. In formulæ if we denote a valid coordinate system by x and another by x' then there is a constant $a \in \boldsymbol{R}$, so that $x' = x + a$. This again, makes it so that distances will agree in both coordinate systems.

Exercise 1. *For two points A and B with coordinates x_A and x_B or x'_A and x'_B, show that the distance between A and B in one coordinate system is the same as it is the other. (Hint: Follow the demonstration above for the analogous fact for time.)*

[2] Mathematicians do not like to use the symbol $=$ for things that equal in a way that is different from how two numbers are equal. In this case they would write $\boldsymbol{T} \cong \boldsymbol{R}$ say that "the space of all times is *isomorphic* to the space of real numbers, but not canonically so. The use of "not canonically" means that there were some choices involved.

Now let's add a dimension: Instead of just forward/backward, we can now also move left/right. The new direction is just like the first and–importantly–they now share a unit: The ruler we defined before will also be used for this additional direction. The two directions are essentially the same *except* that they extend in different "directions". In fact, in 2D, we cannot only measure distance, but also angles (with a protractor).

A *Cartesian coordinate system* for this 2D space consists of the following information:

- a choice of origin
- a choice of "x-axis" taken to be a direction in 2D to be identified with $\boldsymbol{R}$ as above
- a choice of "y-axis" required to lie along the unique direction orthogonal to the first and an identification of it with another copy of $\boldsymbol{R}$.

It is always possible to choose a Cartesian coordinate system for 2D space so, again, up to the choices made, we see that 2D space can be identified with the coordinate plane $\boldsymbol{R}^2$ from coordinate geometry and algebra.

However, this time something is new: Whereas previously two coordinate systems were equivalent only if they differed by a translation, now they can differ by a 2D translation and/or a *rotation*. This is so because translations and rotations both preserve distances and angles.

Adding a third direction proceeds much as before, but there is a complication we glossed over previously: When we added the second direction in the cartesian coordinate system, we didn't say which way it should "point" (*i.e.* which direction orthogonal to the first gets to map to $+1 \in \boldsymbol{R}$). In 3D, this is irrelevant because we can rotate around the x-axis to turn the y-axis up-side down, so to speak.

However, to attach a third axis, we now really do need to decide on yet another convention as to which way it should point. We do so with the following rule: hold your right hand open with out-stretched fingers pointing along the x-axis. Then curl your fingers toward the positive y-direction. While doing this, stick out your thumb: That is the positive z-direction.

It sounds like a joke, but sadly it is not. This is the so-called right-hand rule for completing a Cartesian coordinate system in 3D by assigning the positive z-direction. Such a coordinate system is called a "right-handed coordinate system".

Exercise 2. *Define a left-hand rule, and use it to construct a left-handed Cartesian coordinate system. Can you rotate your new system of coordinates into a right-handed system?*

Our Galilean Space-Time A change of Cartesian coordinate system is called a coordinate transformation. In the examples above we had primed and unprimed coordinates (t, x, y, z) and (t', x', y', z'), and we can imagine transforming one into the other. We have already enumerated these:

Time translation $t \to t' = t + c$ for some $c \in \boldsymbol{R}$. In this case, we keep everything else fixed but just shift all times by a fixed amount. (Kinda like daylight savings time...)

Space translation $x^i \to (x')^i = x^i + a^i$ for $i = 1, 2, 3$ for the three directions and some fixed triplet of numbers $a^i \in \boldsymbol{R}$ for each i.

Rotation In addition, we can rotate the unprimed coordinate system to get a new one. (We won't write the formulæ for this, since it would take us too far outside of the scope of the class.) If we start with a right-handed Cartesian coordinate system, we will get another right-handed Cartesian coordinate system.

The number of parameters we need to specify such a transformation is 7: 3 rotation angles + 3 translations + 1 time translation.

Space and time do not change when we transform coordinates in this way. Because of this, these transformations are often referred to as "symmetries" of space and time, or "spacetime symmetries". In the next section, we will begin to formulate our rules of motion inside this "space-time".

1.2 Particles

We will be talking a lot about particles. By this we will mean a point-like particle, or a particle taking up no volume in space. For the purposes of mechanics, this is an idealized notion, since all physical objects with which we are concerned in classical physics take up some volume in space.

Nevertheless, there are excellent reasons to use this idealized concept. The first is that a physical object can be described very accurately as a collection of point-like objects. If you like, you may think of them as being the positions of the physical atoms out of which they are made up.

A second important reason to talk about point-like particles is that the laws of nature that physical objects obey can often be shown to be equivalent to simpler laws about point particles that inherit certain properties of the original object. For example, the motion of a brick of mass m as it slides off of a roof is equivalent to the motion of a point particle of mass m located at a certain point within the brick called the brick's "center of mass". (We return to this concept in section **??**.)

The position of a point particle is an absolute thing, but to describe it, we need to refer it to something else: To describe where to find your phone, someone must refer to it relative to something you already know how to find. One method to do this is to set up a system of coordinates in which we agree on an origin, and three mutually-independent directions which we usually take to be orthogonal. Even more arbitrarily, we conventionally refer to the distances along these axes by the variable names x, y, and z. The point with coordinates $x = 0$, $y = 0$, and $z = 0$ is called the origin O of the coordinate system. Once we agree on such a system, the position of a point particle can be

described by providing exactly 3 real numbers called the coordinates of the particle with respect to this coordinate system.

The fact that it takes exactly 3 coordinates to describe the position of something is what makes that thing a particle in 3 dimensions. If we were to describe a string, we would need (infinitely) many more points since every point on the string is a point particle. If we were to describe a sheet, we would need even more than that, and so on.

Returning to our coordinate system, we can compare the position of our object to the position of the origin of the system of coordinates. The difference in these positions called the *displacement* from the origin to the position of the object. In our coordinate system, it takes 3 numbers to tell us how far we have to go in each direction in the process of going from the origin to the point.

Displacements are examples of *vectors*. Vectors are objects that satisfy a collection of properties that you can understand from displacements. For example, following a displacement from a position A to another position B with a displacement from B to C gets you from A to C–a third displacement. We represent these displacements as $\mathbf{d}_{AB}$, $\mathbf{d}_{BC}$, and $\mathbf{d}_{AC}$, respectively. We can then write our result of successive displacements as

$$\mathbf{d}_{AB} + \mathbf{d}_{BC} = \mathbf{d}_{AC}$$

But what if we

1.3 Constant Acceleration Kinematic Equations

CAKE 1 When the acceleration is constant

$$\boldsymbol{v}(t) = \boldsymbol{v}_0 + \mathbf{a}_0 t \qquad \Leftrightarrow \qquad \begin{cases} v_x(t) = v_{0,x} + a_{0,x} t \\ v_y(t) = v_{0,y} + a_{0,y} t \end{cases} \tag{1}$$

where $\boldsymbol{v}_0$ is the velocity at $t = 0$.

CAKE 2 When the acceleration is constant, we can find the area under the graph in CAKE 1 to get the displacement as a function of time.

$$\mathbf{r}(t) = \mathbf{r}_0 + \boldsymbol{v}_0 t + \frac{1}{2}\mathbf{a}_0 t^2 \qquad \Leftrightarrow \qquad \begin{cases} x(t) = x_0 + v_{0,x} t + \frac{1}{2} a_{0,x} t^2 \\ y(t) = y_0 + v_{0,y} t + \frac{1}{2} a_{0,y} t^2 \end{cases} \tag{2}$$

where $\mathbf{r}$ is the coordinate "vector".

CAKE 3 It often happens that we need to find t from the first equation and plug that into the second one to solve for displacement. This can be done once and for all by solving CAKE 1 for t and plugging the symbolic expression into the corresponding CAKE 2 equation to get[3]

$$\begin{aligned} v_{f,x}^2 &= v_{0,x}^2 + 2a_{0,x}\Delta x \\ v_{f,y}^2 &= v_{0,y}^2 + 2a_{0,y}\Delta y \end{aligned} \tag{3}$$

1.4 Ballistic motion

In this case

$$\mathbf{a}_0 = \begin{pmatrix} 0 \\ -g \end{pmatrix}$$

with $g = 9.80\ m/s^2$. (In these notes, we will often pretend $g = 10.0\ m/s^2$.) The CAKE equations simplify to constant velocity motion in the horizontal direction

$$\Delta x(t) = v_{0,x}t = v_0 \cos\theta\, t$$

whereas the vertical ones stay non-trivial

$$\begin{aligned} v_y(t) &= v_{0,y} - gt \\ &= v_0 \sin\theta - gt \\ y(t) &= y_0 + v_{0,y}t - \frac{1}{2}gt^2 \\ &= y_0 + v_0 \sin\theta\, t - \frac{1}{2}gt^2 \\ v_{f,y}^2 &= v_{0,y}^2 - 2g\Delta y \\ &= v_0^2 \sin^2\theta - 2g\Delta y \end{aligned}$$

Special case: Can It In this case, a ball rolls off a horizontal table of height h. The time of flight is

$$t_{ToF} = \sqrt{\frac{2h}{g}}$$

which gives a range of

$$\Delta x = v_0\sqrt{\frac{2h}{g}}$$

[3]Nobody ever seems to ask why you can't plug the x-component expression for t from CAKE 1 into the "wrong" CAKE 2 component expression...

Special case: Level range In the most general case of ballistic motion, it is possible to obtain formulae for the time of flight and range, but they are messy and unilluminating because they involve solving a general quadratic equation.

When the final height of a ballistic trajectory is the same as the initial height, however, these formulae simplify: The time of flight reduces to

$$t_{ToF} = \frac{2v_0}{g} \sin\theta$$

Similarly, we obtain the

$$\Delta x = \frac{v_0^2}{g} \sin 2\theta$$

by using the double angle formula, and the maximum height is just

$$\Delta y = \frac{v_0^2}{2g} \sin^2\theta$$

2 Newtonian Dynamics

2.1 Newton I: Inertial frames of reference

We want to be able to describe the laws of nature *quantitatively*. This means that we want to not only say what phenomena will take place (*i.e.* qualitatively), but also how much, how far, how fast in numerical terms. The field of mathematics came into existence precisely to formulate and answer such questions, so we need to establish the correct mathematical framework.

Newton's First Law does precisely this. It asserts the existence of a system of coordinates that is in a state of absolute rest and that this system is the familiar one from geometry.[4] Time is similarly universal, so everyone agrees as to what events are happening simultaneously. This also allows us to compare states of motion in quantitative terms since we can measure angles, distances, and times. The system just described is called an *inertial frame of reference*, *inertial reference frame* (IRF).

Galileo realized that the rules of nature in this inertial frame of reference were not special to this very special frame. The rules would be the same in a frame of reference moving relative to the one that is in a state of absolute rest *as long as that frame was moving at a constant velocity relative to it*.[5] Because of this, we generalize the definition of inertial reference frame to include any and all frames of reference that are moving at constant velocities (sizes *and* directions!) relative to one absolutely at rest. It follows from this that any frame of reference moving at a constant velocity relative to an inertial one is itself inertial.[6]

Newton I Inertial frames of reference are realized in nature, and in such an inertial frame of reference, objects that are free of any interference will continue in their state of motion. This state of motion is to either be at rest or to be moving in a straight line at a constant speed.

Newton also formulated the notions of mass as a quantity of matter and (linear) momentum as a quantity of motion. With these, we can restate the first law as

Newton I (v2) Inertial frames of reference are realized in nature, and in such an inertial frame of reference, the linear momentum of an object does not change in time.

In modern language, we would say that in an isolated system in an inertial reference frame, the linear momentum is conserved.

[4] Mathematically, we say that this is 3-dimensional Euclidean space.

[5] The usual example is to think of a train that is moving at constant speed in a straight line. As long as you don't look outside, you might not notice that you are moving relative to the ground. Of course you can always tell in a real train because of the noises, vibrations, and the fact that the track is not perfectly straight and level. The situation was far worse for Galileo who reportedly tried doing some of his experiments on horseback!

[6] This is an example of what mathematicians would call an equivalence relation.

2.2 Newton II: Forces and motion

Newton's second law is about what forces are and how they act to change an object's state of motion. For us a force will be a push or a pull in some direction with some strength. Multiple forces $\boldsymbol{F}_1, \boldsymbol{F}_2, ...$ can act on a single object, but that single object can only have one state of motion/momentum $\mathbf{p}$. Newton's second law says that the relevant force affecting the object is the vector sum of all the individual forces. This is called the net force:

$$\boldsymbol{F}_{net} = \boldsymbol{F}_1 + \boldsymbol{F}_2 + ... + \boldsymbol{F}_N = \sum_{i=1}^{N} \boldsymbol{F}_i$$

Newton II In an inertial frame of reference, the rate of change of an object's linear momentum $\mathbf{p}$ is given by the net force acting on that object:

$$\boldsymbol{F}_{net} = \frac{\Delta \mathbf{p}}{\Delta t}$$

Note that this means that any violation of the conservation of momentum implies that the system is not isolated and a net force is acting on the object. Conversely, in an IRF, a net force on an object will result in a change of the momentum.

A very important special case is when the mass m of an object is not changing. (This is often what we are thinking of anyway, but it does not always have to be the case.) In this case, the quantity of motion simplifies to

$$\mathbf{p} = m\mathbf{v}$$

where m is the (now unambiguous) mass and $\mathbf{v} = \frac{\Delta \mathbf{x}}{\Delta \mathbf{t}}$ is the velocity. In this situation, Newton's second law simplifies to a more familiar form:

Newton II (v2) In an IRF for an object of mass m, a net force accelerates an object according to

$$\boldsymbol{F}_{net} = m\mathbf{a}$$

Proof: In this case $\Delta \mathbf{p} = \mathbf{p}_f - \mathbf{p}_i = m\mathbf{v}_f - m\mathbf{v}_i = m(\mathbf{v}_f - \mathbf{v}_i) = m\Delta \mathbf{v}$, so $\frac{\Delta \mathbf{p}}{\Delta t} = m\frac{\Delta \mathbf{v}}{\Delta t}$. Since the acceleration $\mathbf{a} = \frac{\Delta \mathbf{v}}{\Delta t}$ is just the rate of change of the velocity, the result follows from the more fundamental formulation of Newton's second law.

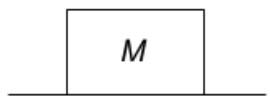
The mass in this formulation acts as a resistance to a change in motion: If we fix the force, but double the mass, the size of the resulting acceleration will be less by a factor of $\frac{1}{2}$. If we triple it, the acceleration is less by a factor of $\frac{1}{3}$, and so on. Very massive objects accelerate very slowly whereas it takes very little force to get a tiny mass to change its velocity. For this reason the mass

appearing on the right-hand side (RHS) of the equation of Newton's second law is sometimes called the *inertial mass*.

Next, we will consider some important examples of the forces appearing on the left-hand side (LHS).

2.2.1 Weight

In the section on kinematics, we have been using that all objects fall at an acceleration of $g \approx 9.8\, m/s^2$ vertically downward. Intuitively, we know that this is happening because of a force pulling the book downward that we refer to as gravity. But what is the size of this force of gravity?



When a book with mass m rests on a table, it is not accelerating downward since the force of the table on the book $\boldsymbol{F}_{tb}$ is keeping it from doing so. By Newton II, a book at rest is experiencing no net force, so the force of the table $\boldsymbol{F}_{tb}$ must be equal in magnitude and opposite in direction to the force of gravity $\boldsymbol{F}_g$:

$$F_{tb} - F_g = ma_y = 0$$

(There are no horizontal forces at all, so we ignore that component of Newton II.) If we somehow suddenly remove the supporting table, the only thing that changes is that there is no F_{tb} term:

$$-F_g = ma_y$$

Since the acceleration is known to be $a_y = -g$, we find that the force of gravity is

$$F_g = mg$$

This combination is called the *weight* of the book. Notice that the weight is *not* a quantity of matter (mass), but a force. Specifically, it is the size of the gravitational force experienced on the surface of the earth by a mass.

Notice that while we were finding this result, we used that the magnitude of the acceleration of a freely falling object was $a = g$. Then the LHS of the equation in Newton II is $F_g = mg$ while the RHS is ma. So when the only force acting on a mass is gravity, Newton II reduces to

$$mg = ma \qquad \Rightarrow \qquad g = a$$

and the mass cancels out.[7]

[7] If something feels like a cheat to you right now, you might be a talented physicist! The forces appearing in this

2.2.2 Normal force

The force of the table on the book above is an example of a *normal force*. These forces are called that because they are pointing in a direction orthogonal to the plane of contact, and another word for an orthogonal angle is "normal".

Later, we will consider books and the like on inclined planes. In these cases, the force of the plane on the book is again normal to the plane. If the plane makes an angle θ with the horizontal, the normal will make the same angle with the vertical.

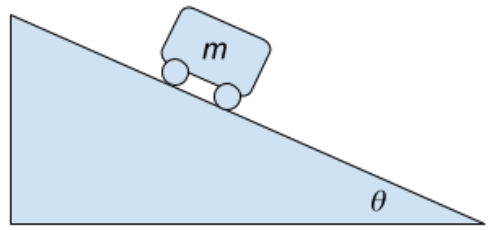


Figure 1: Frictionless cart on an incline

Cart on a ramp Consider a frictionless cart on a table that is inclined with respect to the horizontal by an angle θ. Let's calculate the acceleration of this cart.

The first thing to do is to draw our free-body diagram (FBD). It consists of one dot (the cart) and two vectors emanating from it: the force of gravity $\boldsymbol{F}_g$ pointed straight down and the normal force $\boldsymbol{F}_N$ pointing upward at an angle θ from the vertical and orthogonal to the direction along the inclined plane.

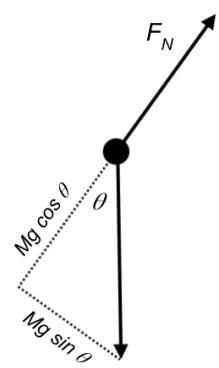


Figure 2: In a free-body diagram for an incline, it is a good idea to decompose in directions along the incline (parallel or ∥) and perpendicular to it (perpendicular or ⊥).

Next we want to apply Newton II to this dot-with-arrows. Since there are only two vectors and they are not pointing in opposite directions, the net force $\boldsymbol{F}_{net} = \boldsymbol{F}_g + \boldsymbol{F}_N$ cannot be 0, so the cart must accelerate. Of course, it will accelerate down the ramp. In particular, it will not accelerate in the normal direction. This means that the normal force must be completely balanced by part of the gravitational force.

To do calculations, we must decompose the forces

section are on the "left-hand side" of Newton II so the mass appearing here "belongs" with those LHS forces. When it comes to the force of gravity, you might call it the "gravitational mass" in the sense that it is related to how strongly the force of gravity acts on it.

In contrast to this, what appeared previously (*i.e.* on the RHS) is the resistance to a change in the state of motion, that is the "inertial mass". Considered in this light, this cancelation greatly bothered Newton who could not figure out any justification for the "inertial" mass having to be *exactly the same thing as* the "gravitational mass"; in principle they could have been two different things. Einstein's theory of gravitation can be said to fix this by making it so that the two concepts cannot be distinguished.

and solve the components of Newton II. Naïvely, we would decompose in the vertical direction and the horizontal one, but this is not the most clever thing to do. In such components, the acceleration will have both vertical and horizontal parts and to find the size of the acceleration in the actual direction of motion, we'd have to use Pythagoras and make a mess.

A much better thing to do is to "rotate our axes": Instead of to the right and upward, we pick "down the ramp" and "orthogonal to the ramp (upward)". When this happens, the acceleration **a** is in the former direction and $\boldsymbol{F}_N$ is pointing along the latter. We do still have the problem that $\boldsymbol{F}_g$ is pointing along neither of these, but now we can just decompose: You should verify that the part along the ramp $F_\parallel = mg\sin\theta$ and that the part orthogonal to the ramp $F_\perp = -mg\cos\theta$. (Note the signs.) The component Newton II equations are now

$$F_N + F_\perp = ma_\perp = 0 \qquad \Rightarrow \qquad F_N - mg\cos\theta = 0$$
$$F_\parallel = ma_\parallel \qquad \Rightarrow \qquad mg\sin\theta = ma_\parallel$$

Notice that the first equation gives us the magnitude of the normal force, and the second gives the acceleration of the cart. Since the only acceleration is $a_\parallel$ we simplify the equations and call this a:

$$F_N = mg\cos\theta \qquad \text{and} \qquad a = g\sin\theta$$

We have derived the equations we sought, but we are not done: The next step is to listen to what they are telling us. The first equation gives the normal force as the weight but adjusted by a trig factor ≤ 1. When $\theta \to 0$, we get the full weight as before. When $\theta \to 90°$, $F_N \to 0$. This makes sense because there is no force keeping a cart on a vertical surface. We also notice that when $\theta \to 0$, there is no acceleration, which is as it should be for a card on a horizontal surface, and when $\theta \to 90°$, $a \to g$ which exactly what we expect for a cart in free-fall.

Finally, for the second equation, we see that the mass has canceled out of the equation for the acceleration. We might think something is wrong with this, but a little reflection and dimensional analysis suffices to convince us that we cannot construct any equation involving the mass. (There are no other variables in the problem with the dimensions of mass, and mass does not appear in the dimensions of acceleration.) *This cancelation from the acceleration is generic.* Do not get confused if an acceleration problem does not give you a mass; it probably means it will cancel out in the end. If you like, you can introduce a mass, use it in calculations, and watch it cancel out in the end.

Exercise 3. *Suppose the angle is $\theta = 30°$ and we do an experiment where the cart starts at the bottom of the ramp and is given an initial speed of $v_0 = 6.0\ m/s$ up the ramp. Calculate how far up the ramp the cart will go. (You may take $g = 10.0\ m/s^2$, for simplicity.) Calculate the time it takes for the cart to get back down to the bottom of the ramp.*

2.2.3 (Non-)Contact forces

Normal forces are examples of a type of force called *contact forces*, because contact is required for the force to exist: Lift the book off the table by picking it up with two hands and there is suddenly no normal force defined. Of course, there are now the forces of your hands acting on the book which are again contact forces.

An example of a non-contact force is the force of gravity. An apple in free-fall on the moon is not in contact with anything.

2.2.4 Friction

Friction is one of the most important forces there is, since there is very little we would be able to do without it. When we walk, we push backward on the ground, and friction with the ground is what pushes us forward. When we slip, we do so for want of friction, and we often fall down. When we fall down, we stop sliding around because of friction, *et cetera*.

Consider again the cart on the inclined plane, but replace it with a book. If the incline is not much, our general experience is that the book will not slide. This is because the force of friction is balancing the parallel component of the gravitational force. We can use this to find the size of the force of friction: As long as the book does not accelerate, the parallel part of Newton II tells us that

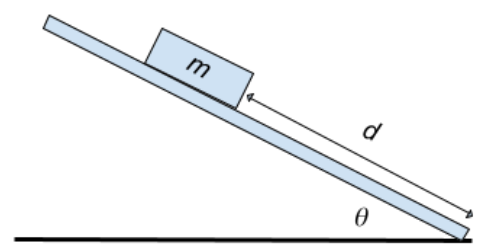


Figure 3: Book of mass m on an incline with angle θ

$$-F_f + mg\sin\theta = ma_{\parallel} = 0 \qquad \Rightarrow \qquad F_f = mg\sin\theta$$

Note that when $\theta \to 0$, $F_f \to 0$. This makes sense because if there were still some non-zero F_f, it would be unbalanced and the book would accelerate from rest. (Think about that: You put the inclined plane with a non-moving book back to horizontal and suddenly the book accelerates to the left?!) Apparently, we need to apply an external force in the direction parallel to the plane in order to have a force of friction. The force of friction will then act to oppose the applied force.

As we increase θ, the applied force increases, and—for a while—so does the force of friction thereby keeping the book from accelerating. Notice that the force of friction is equal in magnitude and opposite in direction while this is happening. Eventually at some angle, the force of friction cannot increase any more beyond some maximum $F_{f,max}$ and the book suddenly accelerates down the ramp. There is still a force of friction, but it is no longer as large as the gravitational component

along the ramp: $F_f < mg\sin\theta$. This component of Newton II now reads

$$-F_f + mg\sin\theta = ma \qquad \Rightarrow \qquad a = g\sin\theta - \frac{F_f}{m} > 0$$

Although friction is still acting, the abruptness with which the change of motion happens suggests that the force of friction for the sliding book is less than the maximum force of friction $F_{f,max}$ for the book at rest. In fact, the acceleration is usually substantial even when the angles just before sliding and just after are almost equal. This can only happen if the force of friction during sliding, is significantly less than the maximum when it is not.

We refine the relevant lingo by referring to the force of friction for a static book as, well, *static friction*. We usually label F_f in this case as $F_{f,s}$ or just F_s. The maximum force of static friction $F_{f,max}$ is similarly re-labled to $F_{fs,max}$ or $F_{s,max}$. The force of friction while the object is moving is referred to as *kinetic friction* and labeled as F_k. The following inequalities hold for the various forces

$$F_s \leq F_{s,max} \qquad \text{and} \qquad F_k < F_{s,max}$$

Coefficients of friction We still have not said anything about how to determine these forces, but we have had so much experience with them, that we might be able to figure it out.

For example, if you are trying to keep something from sliding, one way to do that is to put something heavy on it. In doing so, you are also increasing the normal force. As the excellent physics student that you are, you could design an experiment to find that there is a linear relationship between the maximum force of static friction and the normal force: They are proportional in size. (They cannot be proportional as vectors, because they point in orthogonal directions.) The coefficient of proportionality is called μ (the Greek lowercase letter mu), or because this is for static friction μ_s:

$$F_{s,max} = \mu_s F_N$$

Since $F_{s,max}$ and F_N are both forces, μ must be dimensionless (*i.e.* have no units). The size of the *coefficient of static friction* depends on the type of interface in question. For things sliding on lab tables, a typical value is something like $\mu_s \sim 0.5$.

Exercise 4. *What would it mean for $\mu_s > 1$?*

Encouraged by your success, you might then try to determine the coefficient of kinetic friction using the inclined plane. In this case you might need to figure out a way to determine the magnitude of the acceleration as you vary the mass. If you are successful, you can then use the equations derived above to calculate the force of kinetic friction and you would again find it to be proportional to the normal force:

$$F_k = \mu_k F_N$$

The proportionality constant is now called the *coefficient of kinetic friction*. A typical value in this case might be $\mu_k \sim 0.2$ or 0.3.

The rules for friction are this:

- For a static object, the force of friction is that of static friction F_s. Its magnitude is equal to that of the vector sum of all the forces, but its direction is opposite to it. If the size of the others change, it changes to match so that the net force stays zero. This continues until a limit in the maximum size $F_{s,max}$ is reached. This maximum size is proportional to the normal force on the object as described in the equation above. Note that the size of the force of static friction is always smaller than the max but there is no minimum since it depends on how small/large the remaining forces are: $0 \leq F_s \leq F_{s,max}$.
- For an object in motion (accelerating or otherwise) the force of friction is kinetic. It is again opposite in direction to the sum of the remaining forces, but it always has the same magnitude. Its relation to the normal force is as given in the equation above.
- The constants of proportionality μ_s and μ_k are properties of the materials involved, but they always satisfy $\mu_k < \mu_s$, because the force of kinetic friction is strictly less than the <u>maximum</u> force of static friction.

Exercise 5. *Consider again the book on the inclined plane in figure 3. Show that the maximum angle of the incline at which the book is just barely still in equilibrium is given by*

$$\boxed{\theta_{max} = \arctan(\mu_s)}$$

Reflect, as usual, on your result. (Hint: What is going on physically if the angle exceeds $45°$*?)*

We put a box around this result, because you should memorize it: This, and its inverse relation $\mu_s = \tan(\theta_{max})$ are the ones you need to check quickly whether the system can be kept in equilibrium by static friction alone. (Note: On a test, you may not use this equation without deriving it, so you would do well to remember this exercise.)

2.2.5 Tension, strings, and pulleys

The force of tension is the name for the pulling force along a string. If we are interested in the magnitude, we may refer to the tension to mean the magnitude of the force of tension. However, be warned that care is not always exercised when it comes to treating forces and their magnitudes.

Ideal string A string is considered ideal if it is massless, bends effortlessly and without friction, and does not stretch. Since it is massless, it will not participate in the dynamics of the underlying

problem and there is no FBD for it. Since it does not stretch, the distance between two objects connected by a string does not change. Consequently, the magnitude of the acceleration of one object is equal to that of the other to which it is attached.

Ideal pulley Strings cannot push but they are very useful for redirecting a force. The device that redirects a string is called a *pulley*. Since strings can redirect force, pulleys are used to redirect forces. A pulley that can do this without altering the dynamics of the problem is called an *ideal pulley*. Ideal pulleys have no mass and no frictional forces, so again, they do not appear in an FBD. What this really means in the end is this: *For an ideal pulley, the tension on one side is the same as on the other side.*

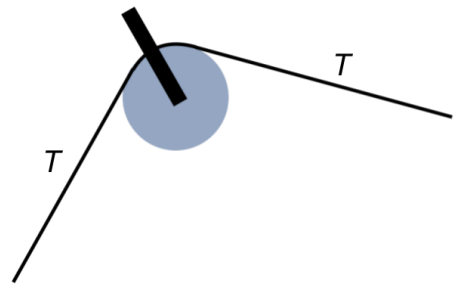


Figure 4: An ideal pulley has equal tensions on the two sides

Engineering application Pulleys are very useful as "force multipliers". Consider the pulley system in figure 5. The mass of the larger grand pianos is about $M = 600\ kg$, corresponding to a weight of $6\ kN$ (~1300 lbs). In this pulley system, there are three cables holding it up, all at tension T, if the pulleys are ideal. If the system is in equilibrium, then the net force on the piano must vanish. For the vertical components, this means that $\sum F_{vert} = Mg - 3T = 0$, or $T = \frac{Mg}{3} = 2\ kN$ (~450 lbs).

The pulley arrangement used here is called a "Luff tackle". It has "ideal mechanical advantage of 3". There are pulley "block and tackle" assemblies with many more cables to further raise the advantage and lower the tension required for equilibrium.

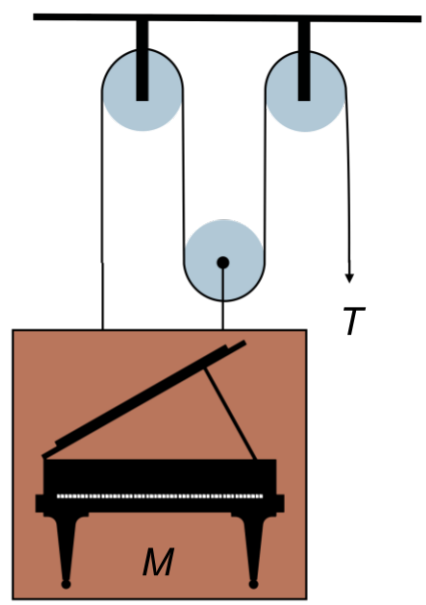


Figure 5: A pulley system called a "Luff tackle" to raise a heavy object

Exercise 6. *Give an example of a system with advantage 1. Design a block and tackle system to give an advantage of 2. Design one to give an advantage of N for any natural number* $N \geqq 1$.

Exercise 7. *After you have finished loading the piano,*

you decide to get in the box and lift yourself. With what fraction of your weight do you have to pull to lift yourself up? (Hint: The answer is not $\frac{1}{3}$.)

2.2.6 Spring force

Ideal strings do not exist. In fact, no physical material can be perfectly rigid and there is always some—perhaps tiny— strechiness to them. Springs are the instrument in physics that can contribute forces by stretching/compressing.

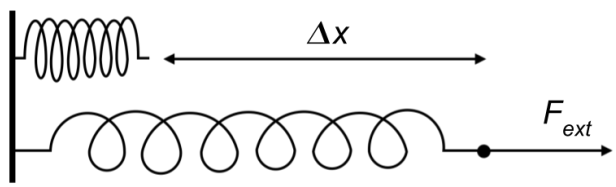


Figure 6: An ideal spring obeys Hooke's rule: It takes an external force $F_{ext} = k\Delta x$ to stretch the spring by an amount Δx. The spring exerts the opposite force $F_s = -k\Delta x$.

A spring is ideal if it is massless, frictionless, and satisfies Hooke's rule: The force F_{sp} of a spring on an object is proportional to the amount Δx that it has been stretched

$$F_s = -k\Delta x \qquad \text{where} \qquad [k] = \frac{M}{T^2}$$

The sign indicates that the direction of the force is *opposite* to the displacement of the spring from its resting length; it is an example of a *restoring force*. Here k is called the *spring constant* and we have explicitly indicated its dimensions. It is a property of the spring. Being ideal, springs do not interfere in the dynamics and do not appear in FBDs.

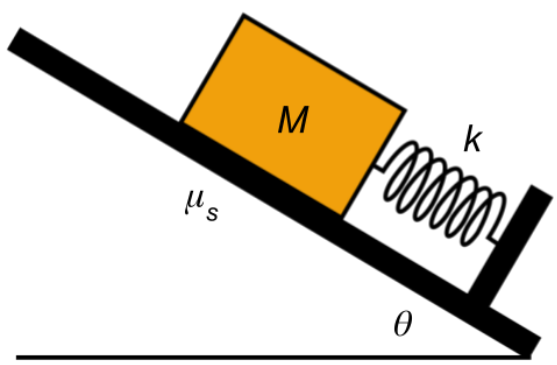


Figure 7: Incline with mass and spring of exercises 8 and 14.

Hooke's rule is often called Hooke's Law, but it should not be, because it is barely a definition. Springs can be ideal in many ways and not be Hookian just because they don't follow the rule to the letter. For example, a spring can be pre-wound in which case a non-zero amount of force is required before it begins to stretch, just like in your spring lab. (After this initial force, it may behave more Hookian.) Other springs are Hookian when stretched ($\Delta x > 0$), but cannot be compressed ($\Delta x < 0$). In problem statements, you should assume a spring is Hookian unless the contrary is explicitly stated. Such springs stretch ($\Delta x > 0$) just as easily as they compress ($\Delta x < k$).

Exercise 8. *A block of mass M sits on a frictionless plane ($\mu = 0$) inclined at an angle θ. It is being kept from sliding further down the incline by a Hookian spring with spring constant k. Show that the compression of the spring is given by the formula*

$$\Delta x = \frac{Mg}{k} \sin\theta$$

2.3 Newton III

Newton's Third Law of dynamics is a statement of reciprocity relating forces acting between two objects. It is often stated as some version of "for every action, there is an equal and opposite reaction". More precisely:

Newton III If an object A acts on another object B with a force $\boldsymbol{F}_{AB}$, then B acts on A with a force $\boldsymbol{F}_{BA}$ and

$$\boldsymbol{F}_{BA} = -\boldsymbol{F}_{AB}$$

For example, if you push on a door, the door pushes back. The magnitude with which the door pushes back is equal to the magnitude of the force you are applying. If you slap a table with your hand, the table slaps your hand back and you feel that as a stinging sensation.

Newton's Third Law may seem innocuous enough, but it is often used in vexing paradoxes of the type: If I push on a cart and the cart pushes back with the same force, how can it accelerate? Don't the forces cancel? The answer to this question is always "No, because the forces are acting on different objects, and forces acting on different objects cannot cancel." When you push on the cart, the cart pushes back on your hand, but that doesn't change the fact that there is an unbalanced force on the cart; the cart will accelerate (provided the force you applied suffices to overcome any friction with the ground, ball bearings, and such). What can also happen though, is that when the cart pushes on you, you may also accelerate in the opposite direction of the cart's acceleration. What is stopping you from doing that is usually friction with the ground: The cart pushes on you, but friction also pushes on you, so if there is enough friction (which is what your rubber soles are providing), you will not accelerate.

If, on the other hand, there is no friction with the floor, say because you are standing on ice, then you will–in fact–accelerate: One way to accelerate yourself when you are standing on ice is to push on something very massive.

Suppose you, an object of mass m, are standing on a frictionless surface. Within reach is a large mass M also on that frictionless surface. Suppose you push the large mass with a force of magnitude F in an attempt to accelerate it away from you. That mass will indeed accelerate away from you in the amount $a_M = \frac{F}{M}$. However, by Newton's Third Law, that mass will apply a force on you of equal magnitude F, so you will accelerate too in the amount $a_m = \frac{F}{m}$. Notice that if you are less massive than the other object, you will accelerate more: $a_m > a_M$.

2.4 FBDs: The Free-Body Diagram Technique

The main tool for analyzing the dynamics of a system using Newton's three laws is the free-body diagram. We first describe the procedure and then demonstrate on some examples.

1. For each mass in the problem, draw a solid dot. We will need to do many things for each dot, so make sure they are well-separated on the page.
2. For each dot, draw an arrow *emanating from the dot* for each force acting on the corresponding mass.
3. For each dot, write a separate copy of Newton's second law formula $\sum_i \boldsymbol{F}_i = m\mathbf{a}$. The mass in this formula is the mass corresponding to the dot on which you are focusing. This is a vector equation, so we will need to reduce it to components in order to work with it.
4. For each dot, choose a convenient system of coordinates and decompose the vector equation into components. For example, you could choose horizontal/vertical or, if an inclined plane is involved, along the plane/perpendicular to the plane. Pick which directions are positive and write the two component equations with all the correct signs. Note that you can choose these coordinates dot-by-dot: You do not need to commit to one system and apply it to every dot.
5. You now have a system of equations. There is a maximum of two equations for each dot and there are as many dots as there are masses in the problem. This gives enough information to solve for all the unknowns, so you should write symbolic equations for the dynamical variables (usually a and T).
6. Reflect on the meaningfulness of your equations and test the limits. Is everything well-defined when some masses are equal? Do you get what you expect when you take masses to be large/small? Can you take angles to special values to recover previously-known formulæ?
7. Plug numbers to get answers to the problem.

Notice that you cannot check whether your answer is sensible in the penultimate step if you were plugging numbers from the start.

A comment about Newton III In the previous section, we formulated Newton's Third Law of dynamics. There we emphasized that the "action" force and "reaction" force act on different objects. What this means for FBDs is that *the two forces of a Newton III pair can never appear on the same dot in the diagram*.

A (long) note about choosing coordinates In step 4, you are told to choose convenient coordinates, but what does that really mean? When there is only one mass in your FBD, it means just what it sounds like, but if there are multiple masses, how should we do it? To answer this question, let us notice something important about the fully labelled FBD with all its dots: Each dot has its own forces and Newton II rules, but some dynamical variables are shared across dots: The acceleration in one part will be related to that in another and/or the tension in appearing for one dot will be the same as that appearing in the another. These variables are what connect the equations across the entire diagram allowing you to solve for them. Because of this, the way you should choose your coordinates should be dictated by whatever makes these connections the simplest.

To fully explain this, we need a complicated set-up. We won't solve it yet, but we will use it to explain what is meant here. Consider the example problem called "Ramp and pulley" in section 2.5.5 below. Figure 14 shows a block of mass M on an inclined plane with angle θ connected by an ideal pulley to a freely-hanging mass m. The two masses appear as separate dots in the FBD shown in figure 15. The question is how to choose coordinates. For the second mass, this is obvious: The dynamics are entirely vertical so we pick a vertical/horizontal decomposition. My preference is to orient things so the accelerations are positive when you cut all wires and turn off all frictions, so for this I would pick downward as the positive direction (but you do you). (It is also my preference to let the simplest part of the FBD determine everything else, and this part is clearly simpler because the horizontal direction is irrelevant, so it reduces to 1D dynamics.)

Next, we look at the larger mass M where things are more complicated since it's on an incline. We could choose horizontal and vertical again, but there is a great reason to *not* do that: the string connecting them. This simple element–with the help of the pulley–connects this part of the FBD to the one for the hanging mass m, and it does so in a way that the acceleration of M is related to that of m. This relation is only direct in the *direction upward along the ramp*. I cannot emphasize enough that these accelerations do not agree as vectors, since they point in different directions. We are cleverly *choosing directions* so that they will match magnitude-wise!

This connection between different parts of the diagram is so strong, that we predicate all consequent choices on it: Imagine what a mess you would make here were you to insist on vertical

& horizontal over along string/ramp & orthogonal to it!

That is how we choose axes.[8]

2.5 Examples

Throughout this section, we will take $g = 10.\ m/s^2$.

2.5.1 Pushing on blocks

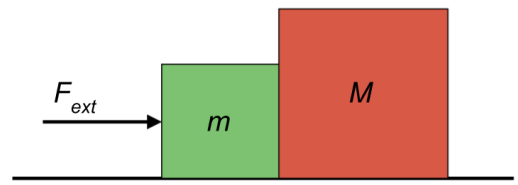


Figure 8: Pushing blocks

Consider an arrangement of blocks at right. We apply a force $F_{ext} = 20\ N$ to the smaller block. The smaller block, in turn, applies a force of magnitude F_{mM} to the larger block. The larger block returns the favor according to Newton III, applying a force of equal magnitude $F_{Mm} = F_{mM}$ in the opposite direction. Let's figure out what happens. We take everything to be frictionless.

The FBD for this problem consists of two dots (the blocks): The forces $F_{N,m}$ and $F_{N,M}$ are

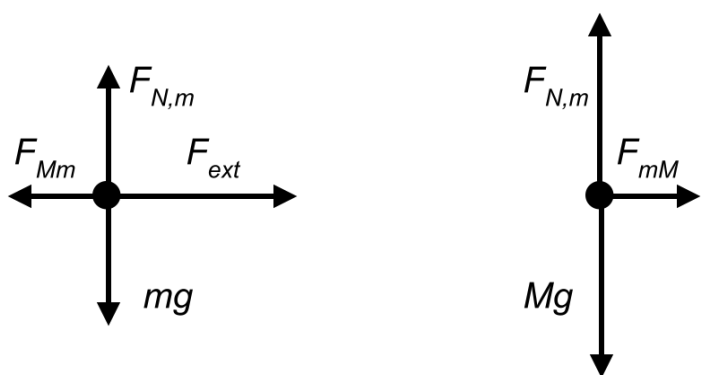


Figure 9: FBD for blocks

the normal forces for the two blocks. Since they are the only ones acting in the vertical direction, applying the vertical component of Newton II tells us that $F_{N,m} = mg$ and $F_{N,M} = Mg$.

[8]Is it consistent? Yes: Each part of the FBD gets its own Newton II equation and its formulation relies only on an IRF. You can think of each part of the FBD as having its own IRF which are all at rest relative to each other but displaced and rotated. Displaced and rotated IRFs are still IRFs! Remember: Everything goes, as long as they are not accelerating relative to each other.

The force F_{mM} applied by the smaller mass acts only the larger one, so it appears in the second part of the FBD only. The reciprocal force F_{Mm} acts only on the smaller mass and appears, therefore, only in the first part of the diagram. These forces are equal in magnitude. The external force F_{ext} acts only on the first mass, so it too appears in that part of the diagram.

Applying the horizontal component of Newton II gives $F_{ext} - F_{Mm} = ma$ and $F_{mM} = Ma$. Together, these give the acceleration as

$$a = \frac{1}{m+M} F_{ext}$$

Plugging this back in, we also find that

$$F_{mM} = F_{Mm} = \frac{M}{m+M} F_{ext}$$

Now we reflect: The formula for the acceleration makes sense because you could really think of the two blocks together as one object of mass $m + M$. The formula for the (re)action force has the property that $F_{mM} \to 0$ when $M \to 0$. In this case, there is no second block, so the force of the first on should vanish. Similarly the second block cannot exert a force on the first if it is not there. A second limiting case has $F_{mM} \to F_{ext}$ when $m \to 0$. This means the first block is not there, so F_{ext} would act directly on the second box. In both cases, the system accelerates as if there is only one box, so the equation for the acceleration has the appropriate limits as well.

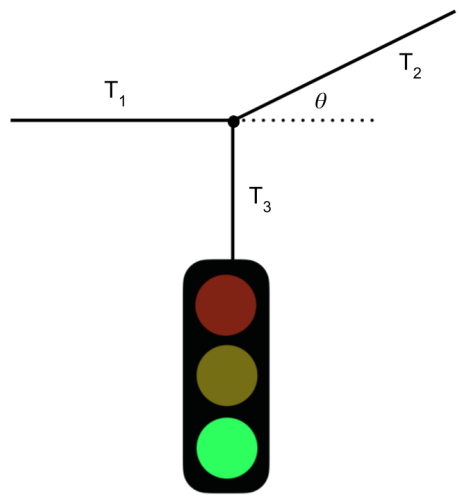


Figure 10: A stoplight hanging from an arrangement of three cables

2.5.2 Hanging mass

Consider the stoplight in figure 10. The light has a mass of $M = 20\ kg$ It hangs from 3 cables with tensions T_i for $i = 1, 2, 3$. The second cable is at an angle $\theta = 30°$ relative to the horizontal. The goal is to determine the tensions. First we will do this in words, and after that, we will organize the work into an algorithm.

The stoplight weighs $F_g = Mg = 200\ N$ and it is in equilibrium. Therefore $T_3 = 200\ N$. The node connecting the three cables is also not changing its position, so the other two cables need to provide a force of 200 N directed upward. Since cable 1 is horizontal, only cable 2 can provide this vertical component. From the geometry, that component is $T_2 \sin\theta$ so the tension

$T_2 = \frac{Mg}{\sin\theta} = 400\ N$. This gives a horizontal component of $T_2 \cos\theta$ that needs to be balanced by cable 1, so $T_1 = T_2 \cos\theta = \frac{Mg}{\tan\theta} = 346\ N$.

2.5.3 Atwood Machine

Consider two masses m and M connected by an ideal string stretched over two ideal pulleys as shown at right. The goal is to determine the acceleration a of the system as well as the tension T in the string. The FBD consists of two dots, each with an arrow for the weight and for the tension. If $M > m$ the mass on the left will accelerate downward. Since the mass on the right is connected by an ideal string (which, therefore, doesn't stretch) so it must accelerate upward at the same rate.

Figure 11: Atwood machine

The dots are disconnected parts of a single FBD, so we may choose this directions to call positive-increasing for each dot. We then write out the components of Newton's second law for each dot keeping careful track of the signs of the terms according to our chosen conventions for what is positive.

In the case just described, Newton's second law is one-dimensional, so we get the following equations: For the large mass,

$$Mg - T = Ma$$

because the + direction is downward. Note the on the RHS, the large mass appears. For the small mass, the upward direction is positive, so two of the three signs are switched:

$$-mg + T = ma$$

The sign on the RHS is correct because m will accelerate upward with rate a at the same time that M will accelerate downward at rate a (since they are connected by an ideal string).

As expected, this has given two equations for two unknowns. If we add the equations together, the tensions cancel, and we are left with

$$Mg - mg = Ma + ma \qquad \Rightarrow \qquad a = \frac{M - m}{M + m} g \tag{4}$$

To solve for the tension, it would be nice to take the difference to cancel the as, but one has a coefficient of m and the other M. We can get rid of these by dividing the first equation by M to

get $Mg - T = Ma \Rightarrow g - \frac{T}{M} = a$ and the second by m to get $-mg + T = ma \Rightarrow -g + \frac{T}{m} = a$ Subtracting gives

$$0 = 2g - \frac{T}{M} - \frac{T}{m} = 2g - \frac{M+m}{mM}T \qquad \Rightarrow \qquad T = 2\frac{mM}{M+m}g \tag{5}$$

Now for the reflection part:

- We notice that in the next-to-last step in solving for a, it appears with the prefactor $m + M$. This makes sense because this is the mass of the whole system, and it is the whole system that accelerates.
- If $M = m$, then a vanishes, as it should. When that happens, the tension balances the weight of each mass, and, indeed, we see that in that case $T \to mg$. This also explains the need for the "2" in the formula for T.
- If $m = 0$, then only M is left, and it accelerates at $a = g$, that is, free fall. Also, $T \to 0$ which makes sense if there is no counter-mass to give the tension.
- The formula for a is anti-symmetric under interchange of the masses. That is, if we switch $M \leftrightarrow m$, then a switches sign. This makes sense because if M turned out to be the smaller mass, then the system would accelerate in the opposite direction. Since we picked a to be positive because we thought the system accelerated the other way, that makes sense too. Conversely, T doesn't care which side is more massive: The ideal pulley guarantees the tension is the same on both sides. This is reflected in the fact that the formula for T is symmetric under exchange of the masses.

2.5.4 Table and pulley

Next, we put one of the masses on a table as shown at right. As always, the first thing to do is imagine the situation. If we set up this experiment and let go of the masses, we might think the big mass doesn't move and the small one doesn't either. However, if we imagine the table is *frictionless*, then that cannot be, because then there is no friction force on M to balance the tension in the string. The system must accelerate with the small mass moving downward and the large one to the right. Since the string is ideal, the distance between the masses cannot change so the two must accelerate by the same amount (albeit in different directions).

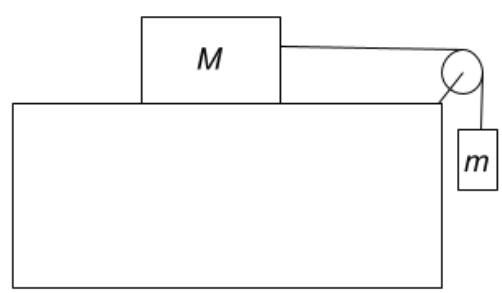


Figure 12: Table with masses connected by an ideal pulley

The free body diagram for this setup has two dots
for M and m, but the former has 4 arrows emanating from the dot (3 if you ignore friction) and the latter has two. Another difference is that the M part is 2D now: We cannot avoid having to split Newton's Law into components.

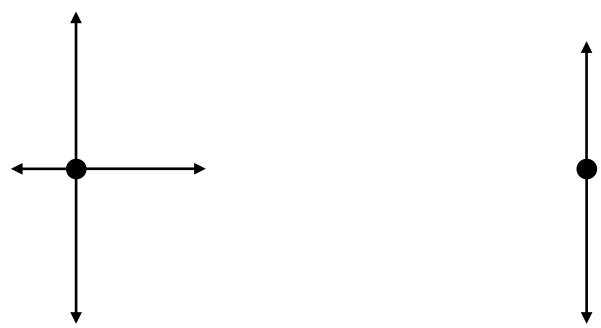


Figure 13: Free-body diagram for table with masses

No friction We begin with the simplified case in which we ignore friction. Focusing on the m dot first, we have only a vertical part to Newton II. Taking downward to be positive for this part of the diagram, Newton II says

$$\sum F_{vert} = ma \qquad \Rightarrow \qquad mg - T = ma$$

Notice that the T arrow upward and the mg downward cannot have the same length as this would imply that m is not accelerating.

For the M part of the diagram, we have to split Newton II into horizontal and vertical parts. The vertical parts are old-hat: There is no acceleration into or off the table because the normal force F_N balances the weight Mg:

$$\sum F_{vert} = ma_{vert} \qquad \Rightarrow \qquad Mg - F_N = 0$$

On the other hand, there is only one force in the horizontal part:

$$\sum F_{hor} = ma_{hor} \qquad \Rightarrow \qquad T = Ma$$

(Of course, we are choosing the positive direction to be to the right.)

This time we have 3 equations for 3 unknowns

$$Mg - F_N = 0 \quad , \quad mg - T = ma \quad , \quad T = Ma$$

Again we have solutions (check this!):

$$F_N = Mg \quad , \quad a = \frac{m}{M+m}g \quad , \quad T = \frac{mM}{M+m}g$$

You should compare these answers to the ones above for the Atwood machine and think about the sensibility of the answers in each case.

The reduced mass If you are working hard to live up to your potential, you have checked these and may have noticed that $m + M$ appeared in front of a again, just as before. You may also have noticed that the combination

$$m_{red} = \frac{mM}{M+m}$$

appeared again. This combination is called the "reduce mass", and it will continue to appear in problems throughout physics. If you are good at algebra and an industrious student, I recommend you show that $m_{red} < m$ and $m_{red} < M$, so it is less than the smaller of the two masses. This also implies that $m_{red} \to 0$ if either mass vanishes. You should also check that $m_{red} \to \frac{m}{2}$ when we take $M \to m$.

Adding friction We will now add friction. Suppose $M = 5.0\ kg$, $m = 3.0\ kg$, and $\mu_s = 0.4$. Also, in these notes $g = 10.0\ m/s^2$, because we are among friends. Now, is this system in static equilibrium?

This sounds like a hard problem, but it is really not difficult to answer, once you understand what to check: We will calculate the force of friction $F_{s,needed}$ that would be necessary for equilibrium, and then we will compare that to the maximum force of static friction $F_{s,max}$. The system will be static if $F_{s,needed} \leq F_{s,max}$, and the force of static friction $F_s = F_{s,needed}$ will be the one we computed. However, if it turns out that $F_{s,needed} > F_{s,max}$, then we would need to exceed the maximum available static force, so the system will not be in equilibrium, and it will begin to accelerate.

Let's now carry out this plan: We suppose for the sake of argument that it is in equilibrium. Then neither mass is accelerating. For the small mass in particular, this means that the tension must balance the weight: $T = 30\ N$. Is this possible? For the large mass on the table, it would mean that the force of static friction is equal to the tension: $F_s = 30\ N$. This could fail to be possible, so let's check. We know the formula for the *maximum* force of static friction $F_{s,max}$. If $F_{s,max} \geqq 30\ N$, everything is fine and the system is in equilibrium.

However, we recall from section 2.2.4 that $F_{s,max} = \mu_s F_N$, so suddenly, we actually care what the normal force is. Since nothing changed in the vertical direction $F_N = Mg$ still, so

$F_{s,max} = \mu_s Mg = 0.4 \times 50\ N = 20\ N$. This is less than the 30 N we need, so the system cannot be in equilibrium.[9]

OK! So we know that the system is accelerating because max static friction is not sufficient to keep it from doing so. The next logical step is to figure out the dynamics: Let's find a and T again!

Actually, we don't even really need to redraw our diagrams. The only things that change are

- Nothing for the small mass m: It was never in equilibrium and still is not. WARNING: The tension changes though, because the dynamics of the system as a whole are different. Friction will "slow down" the acceleration of m, so don't go plugging old values of a and T in for this part.
- The large mass M has a new contributor to its part of the FBD: Newton II becomes $T - F_k = Ma$.
- The force of kinetic friction $F_k = \mu_k F_N$, and the normal force was $F_N = Mg$, so $F_k = \mu_k Mg$.

Putting the last two observations together, we find that $T - \mu_k Mg = Ma$. Combining this with the unchanged equation $mg - T = ma$ for the other mass, we find (verify!)

$$a = \frac{m - \mu_k M}{M + m} g \qquad \text{and} \qquad T = (1 + \mu_k)\frac{mM}{M + m} g \tag{6}$$

Again, we are not done: We have our equations, but the point of deriving them is to *learn/do some physics*. Firstly, if $\mu_k \to 0$, we should recover the answers we already had, and we do. Secondly, we expect friction to reduce the acceleration, and the sign in the acceleration formula implies that it does just that. For the tension it looks like it adds, which also makes sense: more drag; more tension.[10]

2.5.5 Ramp and pulley

As our next complication, let's introduce some non-trivial geometry! Consider the generalization of our previous set-ups in which the table is replaced with an inclined plane.

Let us note from the outset that $\theta = 0$ corresponds to the table in the previous section, and $\theta = 90°$ is the Atwood again. We will start with no friction and then ask the same kinds of questions

[9] This form of argument is called "proof by contradiction": You start by pretending that the wrong answer is true, and you show that that leads to a contraction. In this case, we started by assuming the system was in equilibrium and we concluded that $30 < 20$ which is not true, *i.e.* a contradiction.

[10] Finally, here is a curious thing: If we crank $\mu_k \to 1$, we recover Atwood. Does that make sense??The answer is: In a weird way, yes. Kinetic friction is a curious thing because it "redirects" the normal force to go sideways: To get the sideways frictional force F_k you start with the vertical normal force F_N, take the magnitude, and then multiply by a coefficient (μ_k). So in these equations, if you set $\mu_k \to 1$ it's like setting $F_k \to F_N$ *even though these are not in the same direction*. If you just pretend they are, then these are exactly the forces appearing in Atwood when you "rotate" the friction to be vertical.

as before. After you have derived your answers, you should check that you recover your previous results in the appropriate limits.

Let's set $\theta = 30°$, and let's solve for the dynamics. The FBD is given in figure 15, but to draw it, I had to guess at the direction of the force of friction. Let's see whether I guessed correctly by considering the frictionless case.

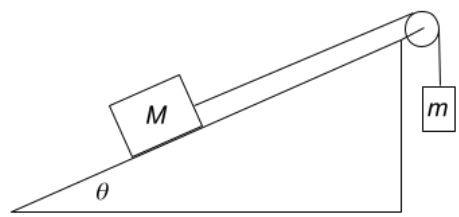


Figure 14: Inclined plane with masses and ideal pulley

The next step in the FBD algorithm is to apply Newton II in the form $\boldsymbol{F}_{net} = m\mathbf{a}$ to each part. For the small mass, the dynamics are non-trivial only in the vertical direction. If we take the downward direction to be positive, the net force is $\sum F_{vert} = mg - T$ and Newton II reduces to

$$mg - T = ma$$

For the large mass, we need to decompose the 2D vector equation into components. We may choose axes different from those for the small mass, *because this is an entirely different part of the diagram*. If we choose to decompose into vertical and horizontal directions, then only one vector (the weight) lines up with an axis. Also, if there is an acceleration up or down the ramp, it would have components in both directions and we would have to do more work.

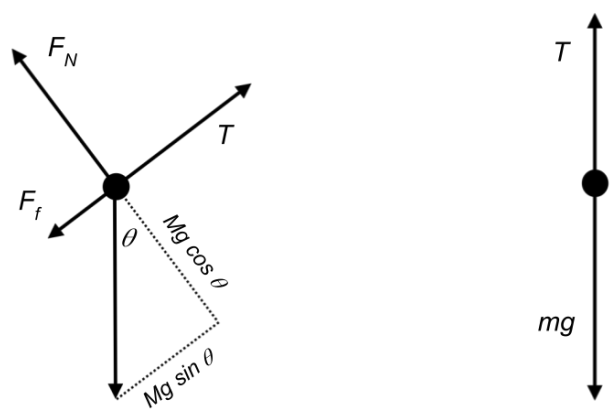


Figure 15: Free-body diagram for inclined plane with masses and pulley

A better choice is to decompose with respect to directions up the ramp (parallel or ||) and orthogonal (perpendicular or ⊥) to the ramp:

$$\sum_i \boldsymbol{F}_i = M\mathbf{a} \longrightarrow \begin{cases} \sum F_{\parallel} = T - Mg\sin\theta = Ma & \text{parallel} \\ \sum F_{\perp} = F_N - Mg\cos\theta = 0 & \text{perpendicular} \end{cases}$$

Note that I'm allowed to use the same T and the same a for both dots because firstly, the pulley is ideal, and secondly the string is ideal (so both masses must have the same acceleration).

The perpendicular equation for M just tells us the normal force is $F_N = Mg\cos\theta$. The parallel equation can be combined with the single vertical equation for m to give

$$a = \frac{m - M\sin\theta}{M+m}g \qquad \text{and} \qquad T = (1+\sin\theta)\frac{mM}{M+m}g$$

,

This was for the frictionless case. For the numbers we decided ($g = 10\ m/s^2$, $M = 5.0\ kg$, $m = 3.0\ kg$, $\theta = 30°$), we find $F_N = 43.\ N$, $a = 0.63\ m/s^2$ and $T = 28.\ N$. Notice that $a > 0$ so the small mass accelerates downward while the large one accelerates up the ramp at the same rate.

Exercise 9. *At which angle is the system in equilibrium?*

When we include friction, this observation implies that the frictional force will be pointing down the ramp, just as we originally guessed. (Had the acceleration been negative, that would have meant that the large mass would slide down the ramp pulling the small mass upward. In that case, the frictional force would have been directed up the ramp.)

Next, we put $\mu_s = 0.4$ and ask: "Is the system in equilibrium?" We prove that it is as follows: If it were not in equilibrium, then that would mean that the maximum possible frictional force

$$F_{s,max} = \mu_s F_N = \mu_s Mg\cos\theta$$

does not suffice to keep the block from sliding. Is that true? Well, plugging in the numbers, we find for this case that $F_{s,max} = 17\ N$.

How much do we need to keep the block in place? The system is static when the sum of the parallel forces vanishes:

$$\sum F_{\parallel} = T - Mg\sin\theta - F_s = Ma = 0 \qquad \Rightarrow \qquad F_s = T - Mg\sin\theta$$

Since m is also static in this situation, $T = mg$, so

$$F_s = mg - Mg\sin\theta$$

Plugging in the numbers, we find $F_s = 5.0\ N$ for the force of static friction that we would need to keep the system balanced. Since this is (comfortably) under the maximum of $17\ N$ we found above, the system is in equilibrium.

Exercise 10. *For what range of angles will this system be in equilibrium? In other words, what are the maximum and minimum angles for which static friction balances out all the forces?*

2.5.6 Additional problems and mixing

The world is our oyster now. We know how to handle all manner of statics and dynamics problems using the free-body diagram technique. Extensions of these problems can be made by mixing in some CAKE. After all, any problem with constant forces will give constant accelerations.

When you attempt the problems below, remember to find the formulæ first and to reflect on your results. When it's time to plug numbers, you're welcome to set $g = 10.\ m/s^2$.

Exercise 11. *The Linches are running late for brunch at Dr. Osborne's house and Dr. Linch is trying to keep some cheery tomatoes from rolling off a plate as the car is accelerating at* $3.0\ m/s^2$ *from a stop.*

a. *At what angle should Dr. Linch hold the plate?*

b. *The dish sitting horizontally on his lap is also threatening to start sliding. What is the coefficient of static friction between the dish and his legs?*

Exercise 12. *Consider again the book on an inclined plane with friction as in figure 3. Let's set some values as follows: The mass* $m = 2.0\ kg$*, the slope is* $\theta = 35°$*, and the coefficients of friction are* $\mu_k = 0.3$ *and* $\mu_s = 0.5$*, respectively. Suppose we start with the book at the bottom of the ramp and we give it an initial velocity* $v_0 = 10.0\ m/s$*.*

a. *Find the distance* d *the book travels up the ramp before coming to rest.*

b. *When the book comes to rest, show that it will just barely not be in static equilibrium.*

c. *How fast is the book going when it gets back down the to bottom of the ramp?*

Exercise 13. *You are sitting on the tarmac at the airport and your captain has just been given clearance to take off. You hear the engines power up, the brakes come off and you begin to accelerate down the runway to your takeoff speed of* $83m/s$*. In the process, you wonder what your acceleration is, so you quickly pull out your trusty compass, hang the earphones they gave you in front of you and measure the angle they make relative to the vertical direction to be roughly* $\theta = 10°$*.*

a. *Find the acceleration of your aircraft.*

b. *How much time will take-off take?*

c. *What is the distance the plane needs to take off?*

Note that you are not in an inertial frame of reference in this problem. If that bothers you, pick the inertial frame to be the one at rest on the ground and consider the problem in that frame.

Exercise 14. *Block of mass* $m = 4.0\ kg$ *can slide on a plank inclined at* $40°$ *with a coefficient of static friction of* $\mu_s = 0.30$*. However, it is being kept from sliding by a Hookian spring with spring constant* $k = 1000\ N/m$*.*

a. *What is the minimum compression of the spring?*

b. *The spring is compressed further by gently pushing on the book and the book is then slowly released again. When the system returns to equilibrium, what is the compression of the spring?*

Exercise 15. *Come up with three problems of your own. Two should involve friction, one should involve a pulley, one should involve an incline, and one should involve a spring. (A single problem could have all these things.) Write solutions for all three problems.*

3 (Linear) Momentum and Energy

3.1 Momentum and Impulse

In the previous section, we mostly used Newton's Second Law in the form

$$\sum_i \boldsymbol{F}_i = m\mathbf{a}$$

but recall that Newton originally formulated this law as relating the left-hand side to a change in the linear momentum (or "quantity of motion" as he put it):

$$\boldsymbol{F}_{net} = \frac{\Delta \boldsymbol{p}}{\Delta t}$$

Rearranging this as

$$\Delta \boldsymbol{p} = \boldsymbol{F}_{net}\, \Delta t$$

we can read it as saying that a change in momentum due to a force $\boldsymbol{F}_{net}$ is proportional to the new force applied and the amount of time over which it acts.

This is quite intuitive: Suppose you want to push a friend in one of the rolly chairs around the commons. To get a maximum change in momentum (from rest to $\boldsymbol{p}_{max}$) you do two things: Push hard and push long. If you push hard but not long enough, $\Delta \boldsymbol{p}$ is not as big as you could make it. Conversely, if another friend who cannot push hard just pushes for a longer time, then $\Delta \boldsymbol{p}$ will can get just as big or bigger.

This combination of force applied and the time over which it is applied is called the *impulse*. For a constant average force $\boldsymbol{F}_{ave}$ acting for a time Δt, the average impulse is[11]

$$\boldsymbol{J}_{ave} = \boldsymbol{F}_{ave}\Delta t$$

Exercise 16. *A tennis player serves the ball perfectly. The ball has mass* $m = 60$ g *and the velocity of the ball after the serve is* $v_f = 120$ *km/hr. The average force of the racket on the ball is* $F_{ave} = 800$ *N.*

1. *Calculate the change in momentum* Δp *before and after the serve.*
2. *Calculate the time* Δt *that the ball is in contact with the racket.*

Exercise 17. *Now suppose we are back to pushing our friends around in chairs. Your push consists of the following parts:*

- *For the first second, you are not pushing.*

[11] The I is taken by the important notion of "current". Ironically, the current density will later we called $\mathbf{j}$.

- *For the next two seconds, you push with a force of* 800*N*
- *For a second after that, you push with a force of* 400*N.*
- *For a next three seconds, you run alongside but don't push.*
- *You pull with a force of* 600 *N for two seconds.*

Let's calculate some impulses:

1. *Graph the force as a function of time for this push.*
2. *Calculate the change in momentum for each of the parts separately.*
3. *Calculate the total change in momentum in this push.*
4. *Relate your calculations to the area under the F-t graph.*

This exercise shows that it makes sense to define the impulse for a force that is not constant in time, and that the impulse with be the area under the force *v.s.* time graph counted with signs. (Recall that this means the areas under the t axis count as negative.)[12]

Definition 1 (Impulse)**.** When a force $\boldsymbol{F}$ (not necessarily constant) acts on an object over a time interval Δt, the *impulse* it imparts on the object is defined to be the (signed) area under the F-t graph. When the force is constant, the impulse is reduced to the product $\boldsymbol{F}\Delta t$ of the force and the time over which it acts.

The relation between this concept and the change in momentum of an object is called the *impulse-momentum theorem*:

Theorem 1 (Impulse-momentum)**.** *When a net force $\boldsymbol{F}_{net}$ acts on an object, the change in momentum $\Delta \boldsymbol{p}$ of that object is given by the net impulse (i.e. that imparted by the net force).*

The proof of this "theorem" is essentially an application of the rules of calculus to Newton's Second Law.

Exercise 18. *Calculate the impulse imparted by gravity on a* 1.0 *kg mass during a free fall from rest lasting* $\Delta t = 2.0$ *s.*

Exercise 19. *When the force applied to an object of mass m is constant, the impulse-momentum theorem reduces to one of the CAKE equations. Identify which equation is relevant, and use it to prove the impulse-momentum theorem in this case.*

[12] If you noticed this sleight of hand, good for you! The force is a vector, so defining the "area under the graph" requires some clarification. In the event that it matters, split the forces into components and apply the area definition of impulse for each component. Since each component $F^i(t)$ (for $i = x, y, z$) will give a different area, we get three different impulse components J^i. We then put the components back together into an impulse vector $\boldsymbol{J}$.

Exercise 20. *A bulldozer pushes a container with constant velocity* $v_0 = 4\ m/s$ *eastward against a force of kinetic friction of* $20\ kN$ *for a total distance of* $\Delta x = 10\ m$ *before stopping. Calculate the impulse the bulldozer imparts on the crate. Why doesn't the crate end up with some non-zero momentum? For the general form of this problem, show the impulse satisfies* $F_x \Delta x = v_0 J$.

3.1.1 Conservation of momentum

Newton's first law states that if there is no force acting on an object in an inertial frame of reference, then the linear momentum of that object will be constant.

Exercise 21. *Imagine that you are floating in outer space with no gravitational or other influences acting on you with some net force. Suppose that you are holding a bowling ball of mass* $m = 7kg$. *Your spacesuit is very bulky and massive, so your mass is* $m = 210kg$. *Suppose you push the bowling ball away from you with a force of* $160\ N$ *in a time* $\Delta t = 1.5\ s$.

1. *Now suppose you have thrown the ball. Calculate the impulse you impart on it.*
2. *According to an observer at rest with respect to you, what are the momenta of you and the ball? What is the sum of these momenta?*
3. *According to the inertial observer, what is the momentum of the ball upon release?*
4. *Calculate the impulse the ball imparts on you.*
5. *According to the inertial observer, what is your momentum upon release?*
6. *According to the inertial observer, what is the change in momentum from before the ball was thrown to after? Why is this happening?*
7. *Calculate the final velocity of you and the ball according to the inertial observer.*

In this problem the final momenta add up to the same amount as the initial momenta did (in this case **0**). This happened because there were no external net forces acting on the ball/astronaut system. There *were* <u>*internal*</u> forces acting on the parts, but by Newton's Third Law, their impulses were equal in size and opposite in direction.

Definition 2 (Isolated system)**.** A collection of objects with no net external forces acting on it is called an *isolated system*.

The conclusion drawn above would hold even if the inertial observer were moving relative to you and the ball at a constant velocity. The only difference is that the momenta would add up to the total mass multiplied by this constant velocity. In general, we have the statement

Theorem 2 (Conservation of (linear) momentum)**.** *In an inertial frame of reference the total linear momentum of an isolated system is conserved.*

By total linear momentum is meant the sum of the momenta of all the parts. By "conserved" is meant that it does not change.

Example 1. *A cart of mass $M = 10\ kg$ rolls on a frictionless track at a constant speed $v_i = 6\frac{m}{s}$ until it collides with a cart of mass $m = 5kg$ and sticks to it. Calculate the final velocity of the stuck-together carts.*

The solution is found by recognizing that the two carts form an isolated system so the total momentum is conserved. Since the second cart is at rest, the initial momentum is $p_i = Mv_i + m \cdot 0 = 60\ \frac{kg \cdot m}{s}$. This must equal the final momentum. After the carts collide, they stick together so whatever the velocity of one of them is, that must also be the velocity of the other. Let's call it v_f. Then the final momentum of the stuck-together carts is $p_f = (M + m)v_f = 15v_f$. Conservation requires $p_f = p_i$, so $15v_f = 60\ \frac{m}{s}$, so $v_f = 4\ \frac{m}{s}$.

This type of collision is called *completely inelastic*.

Exercise 22. *Redo the previous problem for a general, completely inelastic collision. That is, a cart of mass M and initial velocity v_i collides with and sticks to an initially motionless cart of mass m. Show that after the collision, the two move together with final velocity*

$$v_f = \frac{M}{M + m} v_i \tag{7}$$

Take limits to check whether this formula makes sense.

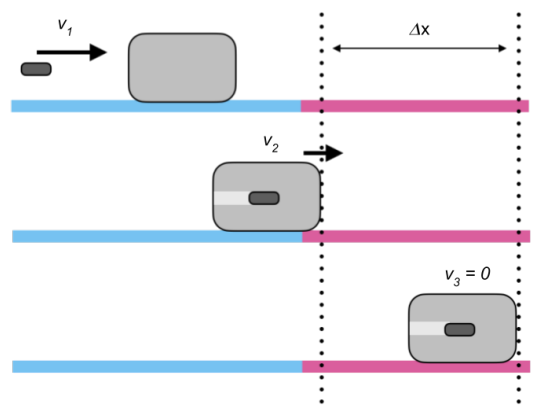


Figure 16: Bullet striking block; a completely inelastic collision

Exercise 23. *A bullet of mass $m = 16\ g$ and speed $360\ m/s$ strikes a block of wood of mass $M = 2.5\ kg$ and gets lodged in it. The block is originally resting on frictionless ice, but after absorbing the bullet, it slides over a rough surface with coefficient of kinetic friction μ_k for a distance of $82\ cm$ before coming to rest.*

1. *Calculate the velocity of the block and bullet just after the bullet becomes lodged in the block.*
2. *Calculate the impulse the bullet imparts to the block.*
3. *Calculate the coefficient of kinetic friction between the block and the rough surface.*

3.1.2 Center of Mass

...

3.2 Work and Energy

In the previous section, we introduced the notion of impulse as the action of a force over a certain amount of time. In this section, we consider the analogous concept for displacement, as with the bulldozer question above.

3.2.1 1 dimension

Exercise 24. *In exercise 19, you connected the concepts of linear momentum and impulse to CAKE 1 in the case of constant net force. Repeat this exercise but in the case of CAKE 3 in 1D. That is, multiply the whole equation by the mass of a particle. Get rid of the factor of two by dividing the whole equation by two. Write your equation so that it looks like*

$$K_f - K_i = F_{net}\Delta x$$

What is the expression for K in terms of m and v?

In this context, the quantity $F_{net}\Delta x$ is called *work* and the quantity K is called *kinetic energy*. Analogously to the conclusion of exercise 19, this result is a special case of the *work-kinetic energy theorem* for constant forces in 1D. Note that in this result, it is important that we are talking about the *net force* on a particle.

Exercise 25. *A model car is attached to a rocket that pushes the car with a force of $F = 7$ N over a distance of* 10 m. *Assuming the car starts from rest and rolls without friction, calculate the final kinetic energy of the car. If the model car and rocket together have mass $m = 2.54$ kg, what is the final speed of the car?*

Again, we will generalize the combination $F\Delta x$ to the area under the F-x curve. In physics, this notion is called "work":

Definition 3. (Work) When plotting the force as a function of x, the *signed* area under the curve between some initial point x_i and a final point x_f is called the *work done by the force* during the displacement from x_i to x_f.

Note that the work done by a force can be negative. This happens when the force and displacement vectors are pointing in opposite directions.

Exercise 26. *Consider pulling your friend across the floor at a constant velocity for a distance of* $3\ m$. *Your friend's mass is* $m = 80\ kg$ *and the coefficient of kinetic friction is* $\mu_k = 0.8$.

1. *Calculate the work done by you on your friend in this process.*
2. *Calculate the work done by friction in this process.*
3. *What are the units of work?*

Exercise 27. *In section 2.2.5, a* $600\ kg$ *piano was lifted using a Luff tackle.*

1. *If the piano was lifted to a height of* $8\ m$, *how much work was done on the piano* by gravity*?*
2. *If the piano was lifted at a constant speed, what was the work done by the person lifting it?*
3. *Recalculate the work done by the person for a general height.*

The units of work in the SI system are called "Joules" and one Joule is one Newton-meter:

$$1\,J = 1\,N \cdot m = 1\,\frac{kg \cdot m^2}{s^2}$$

Exercise 28. *As the excellent young physicist that you are, you would like to know what one Joule of work feels like. Devise an experiment that you could do in the lab to move a mass against the force of gravity so that you do one Joule of work. How big of a displacement and what mass would you pick?*

Note that from the last form given for the unit of Joule, it is clear that the combination $K = \frac{1}{2}mv^2$ has the units of energy, as it should.

Definition 4 (Kinetic Energy)**.** For a particle of mass m and speed v, the combination

$$K = \frac{1}{2}mv^2$$

is defined to be the *kinetic energy* of the particle.

Note that, as the name suggests, this is an energy that is associated to the *motion* of a particle. A trivial, but important consequence of this definition is the following:

$$\boxed{\text{The kinetic energy of a particle is non-negative.}}$$

(This is another way of saying that it is either positive or zero.)

Exercise 29. *Newton originally formulated his laws in terms of momentum* p. *Show that an alternative definition for the kinetic energy is given by the formula*

$$K = \frac{p^2}{2m}$$

Exercise 30. *You compress a Hookian spring with spring constant* $k = 800\ \frac{N}{m}$ *an amout of* $4\ cm$.

1. *How much work did you do? Hint: To answer this question, first draw the F-x graph and calculate the area under the line.*
2. *For a Hookian spring with spring constant k, show that the work done in compressing it an amount x is*
$$W = \frac{1}{2}kx^2$$
3. *This is the work done by you on the spring. How much work does the spring do on you?*

Theorem 3 (Work-Kinetic Energy (1D))**.** *The net work done on an object equals the change in that object's kinetic energy:*
$$\Delta K = W_{net}$$

Exercise 31. *A block of mass* $m = 500\ g$ *sliding at a speed of* $v = 5\ m/s$ *runs into a Hookian spring and comes to rest when the spring is at maximum displacement* $x_{max} = 3.2\ cm$. *Use the work-kinetic energy theorem to calculate the spring constant k. Hint: If you get a confusing sign, consider whether you are computing the work done by the mass on the spring or the work done by the spring on the mass.*

3.2.2 3 dimensions

The relationship you found from CAKE 3 relates the work to the combination of mass and speed called the "kinetic energy". In one dimension, the formula for the kinetic energy is $K = \frac{1}{2}mv^2$, where v was interpreted as the speed of the particle.

Exercise 32. *Generalize the definition of kinetic energy to 3D. Hint: Recall that the velocity* $\boldsymbol{v}$ *is a vector and remind yourself of how we get the magnitude* v *of the vector.*

The kinetic energy shows us the way to generalizing the concept of work to higher dimension.

Exercise 33. *Consider a particle of mass* m *with initial velocity* $\boldsymbol{v}_i$ *and final velocity* $\boldsymbol{v}_f$ *in 3D.*

1. *Write out the change in kinetic energy* $\Delta K = K_f - K_i$ *in terms of the x-, y-, and z-components of* $\boldsymbol{v}_i$ *and* $\boldsymbol{v}_f$.
2. *Apply CAKE 3 for each of the components keeping careful track of x, y, and z directions.*
3. *Collect all the force terms and rewrite them in terms of vectors.*

In summary, when the force is constant, then its components are too, and we have the following statement: If a constant net force $\boldsymbol{F}_{net}$ acts on a particle over a displacement $\Delta\boldsymbol{r}$, then the kinetic energy of the particle is changed according to

$$K_f - K_i = \boldsymbol{F}_{net} \bullet \Delta\boldsymbol{r}$$

This exercise shows us how to define the work in 3D: It consists of the dot product of the force vector and the displacement vector.

Definition 5. When a constant force $\boldsymbol{F}$ is applied to an object that is displaced by $\Delta\boldsymbol{r}$, the work done by $\boldsymbol{F}$ on the object is defined as

$$\begin{aligned} W &= \boldsymbol{F} \bullet \Delta\boldsymbol{r} \\ &= F_x\Delta x + F_y\Delta y + F_z\Delta z \end{aligned}$$

Note that it is crucial that the force be along the direction of displacement: There are no terms in this sum of the form $F_z\Delta x$.

Exercise 34. *You are helping a friend move a couch from their house to a neighbor's house down the street. In the process, your displacement is given in some coordinates by $\Delta x = 20\ m$ and $\Delta y = 50\ m$. Calculate the work done by gravity on the couch in this process. What does this mean for the amount of work you just did?*

Example 2. *Suppose a person of mass $70\ kg$ walks up a flight of stairs a height $h = 4\ m$. How much work does gravity do in this process? If the person is walking at a constant speed, how much work do they do in this process?*

Notice that the horizontal displacement was not given in this problem, but it is irrelevant *since gravity only points downward: The work done by gravity is $W_{grav} = \boldsymbol{F}_g \bullet \Delta\mathbf{r} = mg\Delta y = -mgh$. Since the person is walking at a constant velocity, the net work is zero. Therefore they did $mgh = 70 \times 10 \times 4 = 2.8\ kJ$ of work.*

Again, we will extend the definition of work so that it applies to the appropriate area-under-the-curve concept.

Definition 6. (Work) Consider a force $\boldsymbol{F}$ applied to an object that is displaced by $\Delta\boldsymbol{r}$. The work done by $\boldsymbol{F}$ on the object is defined as the sum of the areas under the respective component graphs of F_x *vs.* Δx, F_y *vs.* Δy, and F_z *vs.* Δz.

Notice that this is done component-by-component. With this understood, the concept of work can be applied to general, changing forces, like the Hooke force.

There are a few important and not-always-intuitive consequences of this definition of work. In the following exercises, we explore a few:

Exercise 35. *You lift a* $100\ kg$ *dumbbell a height of* $h = 1\ m$.

1. *Calculate the work done by gravity on the dumbbell in this process.*
2. *You now put it back down. Calculate the work done by gravity in this second part.*
3. *Calculate the work done by gravity in the process overall, and explain this result in terms of the formula for work.*

Exercise 36. *You push a large box across the floor at a constant velocity against a force of kinetic friction of* $F_k = 10\ N$ *in the following series of steps:*

1. *In the first leg of the circuit, you push the box a distance of* $\Delta x = 10\ m$ *eastward.*
2. *You then turn counter-clockwise* $90°$ *and push with the same force for* $15\ m$ *northward.*
3. *You again turn counter-clockwise* $90°$ *and push with the same force for* $10\ m$ *westward.*
4. *Finally, you turn turn counter-clockwise* $90°$ *and push with the same force for* $15\ m$ *southward, thereby ending up where you started.*

Draw a diagram of your trajectory. Calculate the work done by you on the box and the work done by friction on the box for each leg of the trajectory. Calculate the total work done by you in this process.

Exercise 37. *Now instead of pushing the box around on the floor, you move it in a loop following these steps:*

1. *In the first leg of the circuit, you move the box horizontally a distance of* $\Delta x = 10\ m$ *eastward.*
2. *You then lift the box up* $15\ m$.
3. *Next, you move the box horizontally* $\Delta x = 10\ m$ *westward.*
4. *Finally, you lower the box* $15\ m$ *downward, thereby ending up where you started.*

Draw a diagram of your trajectory. Calculate the work done by gravity the box. Assuming that you did each of these steps with constant velocity, determine the work done by you on the box.

With this, we can formulate the celebrated

Theorem 4 (Work-Kinetic Energy Theorem)**.** *Suppose a net work* $\boldsymbol{F}_{net}$ *acts on a particle. Then the change in the kinetic energy of that particle is given by the net work.*

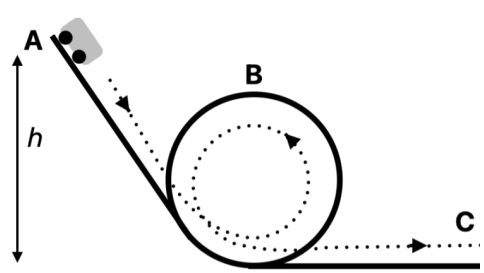


Figure 17: A cart rolling through a loop.

Example 3. *In figure 17, a cart starts from rest at height h rolls down into the loop, rolls up to the top of the loop at B and then out of the loop to C. What is the kinetic energy of the cart at C? What is the speed of the cart at C? Ignore friction.*

Solution: The work done by gravity in the process of going from A to C is $mg\Delta z = mgh$ because the force of gravity only points in the vertical direction. Since this is the only external force acting on the system, we conclude from the work-kinetic energy theorem that this work is equal to the difference in kinetic energies between C and A. Since the kinetic energy at A is zero, it follows that the kinetic energy at C is mgh. To find the speed, we use $K = \frac{1}{2}mv^2$. This gives

$$\boxed{v = \sqrt{2gh}}$$

We put the answer in a box, because this combination appears repeatedly.

3.2.3 Interlude: Review of energy and momentum

Happy New Year!
Let us review what we had been discussing before the break.
We had already defined the linear momentum $\boldsymbol{p}$ which appears in Newton's second law

$$\boldsymbol{F}_{net} = \frac{\Delta \boldsymbol{p}}{\Delta t}$$

For a particle of constant mass m, the momentum can be written as

$$\boldsymbol{p} = m\boldsymbol{v}$$

in terms of the velocity $\boldsymbol{v}$ of the particle. When the net force is constant, we can rearrange the formula to read

$$\Delta \boldsymbol{p} = \boldsymbol{F}_{net}\Delta t = \boldsymbol{J}_{net}$$

where $\boldsymbol{J}_{net} = \boldsymbol{F}_{net}\Delta t$ is called the net impulse. The units of impulse don't have a special name. In SI they can be written as $N \cdot s$ or $\frac{kg\ m}{s}$. In general any number of forces can act on a system and each of them contributes their own impulse, but it is the *net* impulse that results in a change in the momentum of a system. This is the impulse-momentum theorem, and it holds also for non-constant forces if we generalize the impulse to mean the area under the force-time graph.

The impulse momentum theorem for constant forces can also be derived by multiplying CAKE 1 by the mass and rewriting the resulting formula. If instead, we multiply CAKE 3 by the mass and rearrange, we get the analogous formula

$$\Delta K = F_{net}\Delta x = W_{net}$$

where $W_{net} = F_{net}\Delta x$ is called the net work and

$$K = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

is called the kinetic energy of the particle. The unit of energy does get a special name: It is called the Joule. One Joule J is equivalent to one Newton-meter $N \cdot m = \frac{kg\ m^2}{s^2}$.

The relation above is called the work-kinetic energy theorem. Again we can generalize this to non-constant forces and to 3 dimensions. In three dimensions, we need to replace the net work with

$$W = \boldsymbol{F} \bullet \Delta\mathbf{r} = F_x\,\Delta x + F_y\,\Delta y + F_z\,\Delta z = F\Delta r\cos\theta$$

The force $\boldsymbol{F}$ and the displacement $\Delta\mathbf{r}$ are both vectors, and we found that we needed to combine them with the dot product (a.k.a. inner product or scalar product). In the second equality, we have written out this product in components, and in the third we have written it in terms of the magnitudes of the vectors and the cosine of the angle between them.

There are (at least) two crucial things to remember:

1. Only the component of a force along the direction of displacement does work, so when you are carrying groceries down the street in a bag hanging from your arm, gravity is doing no work since the displacement is horizontal.
2. The work that appears in the work-kinetic energy theorem is the *net* work. Plenty of forces can be doing work on a system, but it is the net work that determines whether there is a change in the kinetic energy.

We can use the concept of work to define other forms of energy. Elastic energy is an example and the Hookian spring compressed an amount x can do an amount of work

$$W_{spring} = \frac{1}{2}kx^2$$

where k (in kg/s^2) is the spring constant.

3.2.4 Conservative and Non-conservative Forces

In example 3, we used the work-kinetic energy theorem $W_{net} = \Delta K$ to calculate the speed of a cart that undergoes a complicated roller-coaster motion without having to do any complicated calculus. This is pretty great, but it would be even nicer if we could somehow replace the net work with the change of some function, say $-\Delta U$, because then we get $-\Delta U = \Delta K$ and the combination $K + U$ would be unchanged in the process. In fact, we can do that in this problem because we can take the

function U to be mgh. Then $K_i + U_i = mgh$ and $K_f + U_f = \frac{1}{2}mv_f^2$, and we get the solution we derived.

This *almost* works, but exercise 36 showed us that, in general, the work done by a force in some process depends on the path taken in that process: The work done in that exercise adds up to 500 J even though we went in a loop. The overall displacement added up to 0, but we are adding force times displacement and that came out to $100\,J + 150\,J + 100\,J + 150\,J = 500\,J$. For every part we could compute, $\boldsymbol{F} \bullet \Delta\mathbf{r}$ had the same sign and nothing canceled.

In contrast to this, the work done by gravity vanishes for any process in which the final and initial points have the same height. In exercise 37, you repeated the loop of exercise 36, but got that the answer was 0 J. This time, the signs in $\boldsymbol{F} \bullet \Delta\mathbf{r}$ were different and the balance worked out to $0\,J - 150\,J + 0\,J + 150\,J = 0\,J$ for the work done by gravity. The 0s come about because the force of gravity was orthogonal to the displacement along these legs and the relative sign appeared because gravity always points down and the displacements in these legs were opposite.

Thus work is generally a function of the path taken, but for some forces it does not actually depend on the path. We codify this distinction with a definition:

Definition 7. A force is called *conservative* if the work done by it in any process is independent of the path taken. A force that is not conservative is called *non-conservative*.

Notice that this is a bit meta: We are classifying forces by defining the work $W(\mathcal{P})$ that force does along a path $\mathcal{P}$ and then checking whether it changes if we change the path to another path $\mathcal{P}'$: For any pair of paths $\mathcal{P}$ and $\mathcal{P}'$

$$\begin{aligned} W(\mathcal{P}') &= W(\mathcal{P}) && \text{force is conservative} \\ W(\mathcal{P}') &\neq W(\mathcal{P}) && \text{force is non-conservative} \end{aligned}$$

Exercise 38. *Argue that an equivalent definition would be to say that the force is conservative if the work around any closed loop is zero.*

We make this distinction because the function U that we want exists for conservative forces, but does not for non-conservative forces:

Theorem 5. *Let $\boldsymbol{F}$ be a conservative force acting on an object that is displaced from a point p_i to another point p_f. Then there is a function U depending only on position for which the work done by $\boldsymbol{F}$ can be written as*

$$W = U(p_i) - U(p_f) = -\Delta U$$

That is, the work is the difference between the value of U at the initial point and that at the final point.

Proof. [13] Pick any random point p_* you like–call it the base point–and define a function $U(p)$ to be the work the force $\boldsymbol{F}$ does in going from p to p_*. Since the force is conservative, it doesn't matter how you get from p to p_*, so this gives an honest function for every point p. (Below we will check that it is OK that this definition of U depends on the base point.)

Now consider the work done in going from p_i to p_f. Since the force is conservative, we may as well start from p_i and go to the base point p_* before continuing on from p_* to p_f. Now the work done in going from p_* to p_f is equal in magnitude and opposite in sign from the work done in going from p_f to p_*. (If you doubt this, do both processes successively to end up with a loop. Since the force is conservative the sum is 0 so each part cancels the other.) Thus the work done in this process is precisely $W = U(p_i) - U(p_f)$.

Finally, we should make sure that this process does not depend on the random base point p_* we picked. To do so, suppose we start over again with a different base point p'_* and repeat all the steps with the new function U'. Then $U'(p)$ differs from $U(p)$ by a constant c that is the work done by the force in going from p'_* to p_*: $U'(p) = U(p) + c$. But this means that the work done in going from the initial point to the final point as calculated with this new function U' will be

$$W' = U'(p_i) - U'(p_f) = U(p_i) + c - U(p_f) - c = U(p_i) - U(p_f) = W$$

But this is the same as what we had before, so we have shown that the work done by the conservative force in going from p_i to p_f does not depend on p_* or p'_*. □

This proof actually shows that there is an infinite number of possible functions U that all differ from one another by constant amounts. This ambiguity in the definition of the function corresponds precisely to the freedom of choosing the arbitrary base point used to define the function.

Definition 8. A function related to the work W done by a conservative force as per the theorem above is called a *potential energy function*. Such functions are determined only up to an additive constant which cancels out of the defining relation

$$W = -\Delta U$$

because $\Delta U = U_f - U_i$ does not change if we shift U by a constant.

Exercise 39. *Consider a book of mass $m = 2$ kg resting on a bench of height $h = 90$ cm.*

1. *Construct a potential energy function U_g for the work done by the gravitational force in raising the book a distance z above the table.*

[13] Please feel free to skip over this proof. Although we will use the result extensively, understanding why the claim is true is not crucial to using it.

2. *Demonstrate that $U_g' = U_g + c$ is also a valid potential energy function for any constant c.*
3. *Interpret the meaning of c. Hint: What choices did you make when you defined U_g?*

Exercise 40. *Is the Hooke force conservative? If not, explain why not. If it is, give a potential energy function for it.*

To recapitulate, any type of force can be classified as either conservative or non-conservative. All conservative forces have potential energy functions, whereas non-conservative forces cannot have such functions because they depend on paths instead of just starting and ending points.

The Work-Energy Theorem We will now use all of this machinery to improve on our existing work-kinetic energy theorem.

Since any force is either conservative or not, the net work done W_{net} in any process can be split into conservative W_C and non-conservative W_{NC} parts, and the conservative parts can be written in terms of a potential energy function $W_C = -\Delta U$ by theorem 5. Writing this in formulæ,

$$W_{net} = W_{NC} + W_C = W_{NC} - \Delta U$$

On the other hand, the work-kinetic energy theorem (theorem 4) relates the LHS to the change in kinetic energy $W_{net} = \Delta K$. Therefore, the non-conservative work changes the sum of the kinetic and potential energies:

$$W_{NC} = \Delta K + \Delta U = \Delta(K + U)$$

The combination $K + U$ we were looking for will be important, so we give it a name:

Definition 9. The sum of the kinetic and potential energies is called the *mechanical energy E*.

$$E = K + \sum_i U_i$$

Note that the potential energy $U = \sum_i U_i$ may itself be a sum if there is more than one conservative force acting.

Example 4. *Consider a mass at rest hanging from a stand by a Hookian spring. Suppose we pull down lightly on the mass and release it. Then the mass will oscillate up and down in a regular manner as represented in figure 18.*

1. *Write down the names of the conservative and non-conservative forces acting in this example.*
2. *Suppose that the effects of the non-conservative forces can be ignored, and construct a potential energy function for this mass/spring system. Which choices are you making here?*
3. *What is the mechanical energy just before the mass was released?*
4. *What is the mechanical energy when the mass has its maximum speed?*

In general the non-conservative forces cannot be ignored, but we can still rewrite our work-kinetic energy theorem in a useful way:

Theorem 6 (Work-Energy Theorem)**.** *The change in mechanical energy $E = K + \sum_i U_i$ in any physical process for any system is given by the work done by all the non-conservative forces acting on the system:*

$$W_{NC} = \Delta E$$

Proof. This is just an interpretation, summary, and rewriting of the results we derived above. □

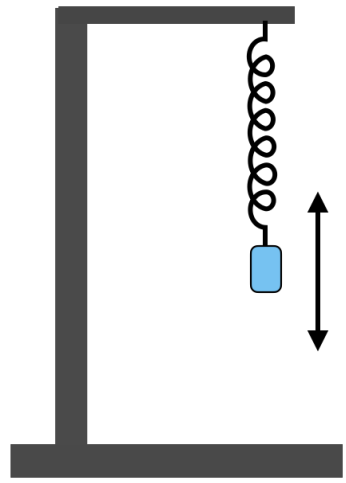


Figure 18: Mass and spring oscillation

Corollary 1. *In a system in which all forces are conservative, the mechanical energy is conserved. In other words, in any process in which there are no non-conservative forces, the sum of the kinetic and all the potential energies in a system is the same initially as it is afterwards.*

Exercise 41. *Go back to example 3, and re-solve the problem using the improved work-energy theorem. Suppose you do this experiment and you find a speed that is lower than what you calculated by a factor of* 0.95*. Use the work-energy theorem to determine the fraction of mechanical energy lost to non-conservative forces. Explain in as much detail as you can what is happening to this energy.*

Example 5. *A ball of mass $m = 100$ g in a launching tube is compressing a spring with spring constant $k = 1000\ \frac{N}{m}$ an amount $d = 10$ cm. The ball is released and it shoots vertically out of the tube as sketched in figure 19. Ignore friction throughout.*

1. *Find the maximum height h that the ball reaches.*
2. *Find the velocity of the ball at the moment it loses contact with the spring.*
3. *Suppose the ball falls straight down landing in the launcher and compressing the spring. What is the max compression of the spring?*
4. *Find an expression for the velocity of the ball as a function of height z when the ball is out of the tube.*

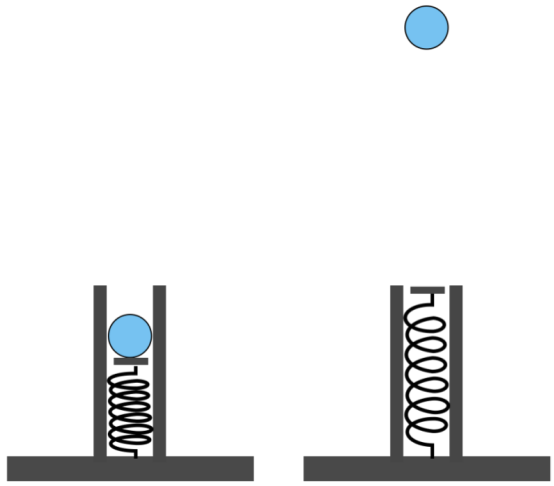


Figure 19: Mass and spring launcher

Solution:

1. We will solve this by saying that the mechanical energy before release is equal to that at the apex. However, to write an energy function, we must "choose the zero" corresponding the to base-point ambiguity of the potential energy. It seems convenient to choose the height $z = 0$ to be the position of the release point, and we may as well take the gravitational potential energy to be zero there. Since we have not released it yet, the kinetic energy is zero, so the only other energy to consider is the spring potential energy. Since the compression of the spring is d, the initial mechanical energy is $E_i = \frac{1}{2}kd^2 = 5\,J$. The total mechanical energy is equal to this throughout the process. At any point in time, the mechanical energy function can be complicated or simple but in any case it must equal $E_i = 5\,J$. At the highest point, the mechanical energy is simple again: The kinetic energy is zero and there is no spring force so $E_f = mgh$, the gravitational potential energy only. Thus

$$h = \frac{kd^2}{2mg}$$

   or $h = 5\,m$ if g $= 10\,m/s^2$.

2. When the ball loses contact with the spring, it has only kinetic and gravitational energy so $E_f = \frac{1}{2}mv^2 + mgz$. When the spring is no longer pushing, it has displaced an amount $z = d$, so $E_f = \frac{1}{2}mv^2 + mgd$. This is what must equal the initial mechanical energy, so $\frac{1}{2}mv^2 + mgd = \frac{1}{2}kd^2$. At this point, you can start plugging numbers, but I'll solve for v:

$$v^2 + 2gd = \frac{k}{m}d^2 \qquad \Rightarrow \qquad v = \sqrt{\frac{k}{m}d^2 - 2gd}$$

Plugging in numbers, this gives $v \approx 9.9\ m/s$.

3. 10 cm. This is because the final energy must be the same as the initial, and this can only be true if the system returns to its original configuration. (Confirm this!)
4. When the ball is out of the tube, there is no spring energy, so the mechanical energy is $E = \frac{1}{2}mv^2 + mgz$. Since this must equal $E_i = \frac{1}{2}kd^2$ at all times, we can again solve for v. The only difference from what we had before is the z-dependence of the gravitational part, so

$$v = \sqrt{\frac{k}{m}d^2 - 2gz}$$

If we want to check this, we can note that z cannot be larger than $z_{max} = \frac{kd^2}{2gm}$, which we recognize as our max height h.

Note that throughout this process, we are continually trading kinetic, spring potential, and gravitational potential energy, all the while keeping the sum equal to 5 J.

3.2.5 Principle of Conservation of Energy

Conservative forces are special in that the work done by them can be written as a change in a function U the potential energy function associated to that force: $W_C = -\Delta U$. Being a function, the potential energy only depends on points, so the work done will be a difference in the value of this functions at the starting and ending point of whatever process we are considering.

Non-conservative forces do not have such a function, because they depend on the entire path along which a force acts, and not just on the end points of the path. Because functions depend on points, non-conservative forces cannot give rise to work that is the difference of a function between two points. Nevertheless, it is possible to write the work done by non-conservative forces as the difference in energy before and after the process. This energy will not be mechanical energy. Instead, it will be the energy associated to some internal change in the system; we call it internal energy:

$$W_{NC} = -\Delta E_{int} = E_{int,i} - E_{int,f}$$

We emphasize that $E_{int,i}$ and $E_{int,f}$ are constants with the units of Joules, *they are not functions*.

The use of introducing this concept is that now we can re-write the work-energy theorem as the conservation of total energy:

Theorem 7. *The total energy of an isolated system is conserved.*

Proof. Split the forces as conservative and non-conservative as before, but write the non-conservative part as above. Then the work-energy theorem becomes $\Delta E_{int} = -W_{NC} = -\Delta K - \Delta U$. Moving all terms to the right, we can rearrange this to $\Delta(K + U + E_{int}) = 0$. Therefore, the total energy $K + U + E_{int}$ is constant. □

This statement is often referred to as the Principle of Conservation of Energy. It states that energy cannot be created or destroyed. Rather, energy can only be changed from one form to another. Examples of the form the different parts of the energy can take are gravitational, elastic (*e.g.* spring), kinetic, thermal, nuclear, electric, and solar. This is not an exhaustive list, and it is possible that we do not yet know all the forms that energy can take.

It is also important to understand that not all forms of energy are interchangeable. For example, the work done by friction often results in a change in thermal energy: Rubbing your hands together makes them feel warm. That thermal energy cannot then be used to compress a spring. Heat is the name for changes in thermal energy, and there is a limit to how much useful work can be extracted from heat. The study of heat and its relation to mechanical energy is the theory of thermodynamics.

3.2.6 Examples and problems

Example 6. *A frictionless cart of mass m sits on a ramp with angle θ from the horizontal compressing a spring with spring constant k and amount ℓ and is subsequently released. It shoots up the ramp a distance d.*

1. *Calculate d when friction can be ignored.*
2. *Now the cart is replaced with a block that slides with a coefficient of kinetic friction μ_k. Recalculate d.*

Solution. 1. *Before the cart is released, the mechanical energy is purely the potential in the spring, if we take the gravitational energy to be zero here: $E_i = \frac{1}{2}k\ell^2$. When the cart reaches its highest point, all the energy is gravitational: $E_f = mgh = mgd\sin\theta$. Setting these equal to each other and solving for d,*

$$mgd\sin\theta = \frac{1}{2}k\ell^2 \qquad \Rightarrow \qquad d = \frac{k\ell^2}{2mg\sin\theta}$$

Note that $k\ell$ has the units of force and so does mg, so this equation makes sense dimensionally. Note also that when $\theta = 90°$, we recover the result from example 5.

2. *When there is friction, the work it does results in a change in mechanical energy, according to the work-energy theorem. Since the work done by friction is $W_{f,k} = -F_k d = -\mu_k mg\cos\theta d$, we find that $-\mu_k mg\cos\theta d = W_{f,k} = E_f - E_i = mgd\sin\theta - \frac{1}{2}k\ell^2$ Rearranging and solving for d gives*

$$\frac{1}{2}k\ell^2 = (\sin\theta + \mu_k\cos\theta)mgd \qquad \Rightarrow \qquad d = \frac{1}{(\sin\theta + \mu_k\cos\theta)}\frac{k\ell^2}{2mg}$$

When we turn off friction, we recover our previous result. On the other hand, as friction increases, d decreases. We cannot compare the $\theta \to 0$ limit with a previous result, because

our dynamics techniques did not suffice to calculate this behavior without resorting to the use of calculus.

Exercise 42. *A frictionless cart of mass m sits on a ramp with angle θ from the horizontal compressing a spring with spring constant k and amount ℓ and is subsequently released. It shoots up the ramp and flies off continuing ballistically until it lands at the same height as the top of the ramp. Ignore friction.*

1. *Calculate the speed of the cart as it leaves the ramp.*
2. *Calculate the horizontal range.*
3. *Calculate the maximum height.*

Hint: First use energy methods to do part 1. Compare your answer with that in example 5. After that you can use your ballistics methods from CAKE. Don't forget to use the appropriate components for the velocity.

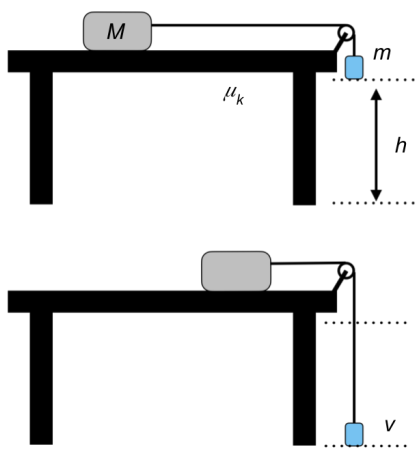


Figure 20: The mass M is connected by a string to a hanging mass m. When the system is released, the small mass falls dragging the large mass a distance h against kinetic friction with coefficient μ_k.

During fourth period, I tried to answer a question from another packet, but in my haste, I made a mistake. It's a good problem though, so I thought I should include it here:

Example 7. *Consider the experiment shown in figure 20. It represents a mass M on a table with coefficient of kinetic friction μ_k connected to a hanging mass m. When the system is released, it accelerates as the hanging mass falls a distance h. Find the velocity of the hanging mass just before it hits the ground.*

Solution *We will use the work-kinetic energy theorem. The net work done by the system has two parts: The work due to gravity acting on the small mass* $W_g = mgh$ *and the work due to friction acting on the large mass* $W_{f,k} = -F_k h$.[14] *The displacement for both of these is* h *since the blocks are connected by a string. The net work*

$$W_{net} = (m - \mu_k M)gh$$

must equal the change in kinetic energy of the system. Since the system we are considering started from rest ($K_i = 0$) *and consists of both masses together,* $\Delta K = \frac{1}{2}(m+M)v^2$. *Setting these equal* $\frac{1}{2}(m+M)v^2 = (m - \mu_k M)gh$ *and simplifying,*

$$v^2 = \frac{m - \mu_k M}{m + M} 2gh \qquad \Rightarrow \qquad v = \sqrt{\frac{m - \mu_k M}{m + M}}\sqrt{2gh}$$

We can confirm this from our work in dynamics, because this is the same setup as the one that gave the results (6). *Substituting that result for the acceleration into CAKE 3 in the form* $v = \sqrt{2ah}$, *we recover our result.*

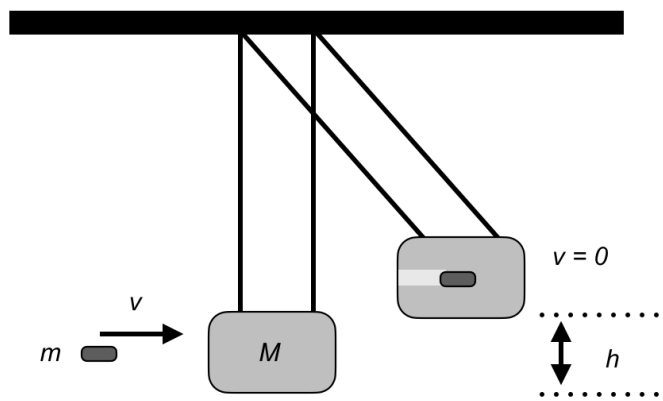


Figure 21: The ballistic pendulum: A bullet of mass m and speed v strikes a block of mass M that is at rest and becomes lodged in it in a completely inelastic collision. The bullet and block system are raised a vertical distance h before momentarily coming to rest.

Example 8. *Figure 21 shows the working of a ballistic pendulum. In order to measure the speed* v *of a bullet of known mass* m, *you can shoot it into a block of known mass* M *hanging on strings. The bullet becomes embedded in the block and they both swing up to some height* h *before coming to rest. Find an expression for the bullet's initial speed in terms of the other relevant data. (Instead of* h *you might instead know the length* ℓ *of the strings and you would measure the maximum angle* θ *from the vertical that the deflection causes. These would be related by* $h = \ell(1 - \cos\theta)$.*)*

[14] For those of you in period 4, I had made the mistake of not including the work done by gravity.

Solution. *This is a 2-part problem: The first part consists of a completely inelastic collision of the type you solved in exercise 23, and the second part is a conservation of mechanical energy problem relating the kinetic energy of the bullet-and-block after the collision to the gravitational potential energy at the highest point.*

You should recalculate that the velocity of the bullet and block after the collision is $v' = \frac{m}{m+M}v$ *(see ex.23). This is used as the initial velocity in the second stage of the problem. At lowest point in the swing, we may take the mechanical energy to be purely kinetic* $E_i = \frac{1}{2}(m+M)v'^2$. *At the highest point, it will be purely (gravitational) potential* $E_f = (m+M)gh$. *Setting these equal and removing a common factor, we get* $v' = \sqrt{2gh}$, *which we recognize from example 3. Setting this equal to the answer from the first stage and solving, we find*

$$\frac{m}{m+M}v = \sqrt{2gh} \qquad \Rightarrow \qquad v = \left(1 + \frac{M}{m}\right)\sqrt{2gh}$$

4 Midterm Review

Recall that you may not use formulas that you did not derive on FRQs. However, this test is entirely MCQs, so some formulae are good to keep in mind. For example, when a ball is thrown, the maximum height and the horizontal range are both proportional to $\frac{v_0^2}{g}$ (see section 1.4), so questions about what happens if you throw twice as hard are easily answered.

- Kinematics
  - xt-, vt-, and at-diagrams
  - relationship between x, v, and a in terms of slopes and areas
  - vector addition and relative velocity problems
  - CAKE 1, 2, 3
  - ToF, max height, horizontal range
  - Change in height, distance, time, *et cetera* when something is doubled, halved, *et cetera*.
- Forces
  - Free-body diagrams and Newton II
  - Recall statics experiments with pulleys and spring scales
  - Forces (and FBD) on inclined plane (together with initial motion up and down the plane)
  - Scales and elevators
  - Pulleys
- Momentum and Energy

- Change of momentum and energies when masses, velocities, distances, *et cetera* are doubled, halved, *et cetera*.
- Change in momentum with $\boldsymbol{v}_f = -\boldsymbol{v}_i$ is not $\mathbf{0}$.
- Momentum and *kinetic* energy are changed only by net force according to the Impulse-Momentum Theorem and the Work-KE theorem, however...
- Graphical determination of impulse and work
- Momentum conservation for isolated system
- *Mechanical* energy is conserved if there are no non-conservative forces (like friction).
- *Total* energy is always conserved but mechanical energy will be transformed into internal/thermal energy if non-conservative forces are present.

5 Rotational Motion

5.1 Circular Motion

Consider an object moving in a circle with constant speed v. Its velocity vector $\boldsymbol{v}$ is tangential to the circle in the direction of motion. It has constant length v, but its direction is continually changing. Because of this, the acceleration $\mathbf{a} = \frac{\Delta \boldsymbol{v}}{\Delta t}$ is non-zero even though the object is not speeding up or slowing down.

Consider the difference in the velocity vector between an initial time t_i and $t_f = t_i + \Delta t$. The speeds $v_f = v_i$ but $\boldsymbol{v}_f$ is rotated slightly relative to $\boldsymbol{v}_i$. Taking the difference $\Delta \boldsymbol{v} = \boldsymbol{v}_f + (-\boldsymbol{v}_i)$, we see that it points radially inward. Since it does not matter which t_i we started with, the acceleration points radially inward at all times. Such an acceleration is called *centripetal*, meaning "toward the center".

To get the magnitude, we note that the triangle formed by $\boldsymbol{v}_i$, $\boldsymbol{v}_f$, and $\Delta \boldsymbol{v}$ is an isosceles triangle with sides v and Δv. This triangle is formed by sweeping $\boldsymbol{v}_i$ through an angle $\Delta\theta$. A similar triangle is formed by sweeping the radius vector through the angle $\Delta\theta$. It has equal sides of length r and a base that is very close to the arc $\Delta\ell = r\Delta\theta$. If the angle is very small, we may approximate this base by $\Delta\ell$. Since space and velocity triangles are similar, the sides satisfy

$$\frac{\Delta v}{v} = \frac{\Delta \ell}{r} \qquad \Rightarrow \qquad \Delta v = \frac{v}{r}\Delta\ell$$

Since we are considering circular motion, r is constant in time. Since the motion is uniform v is constant too. This means that the only thing changing on the RHS is $\Delta\ell$. Therefore, the acceleration has magnitude

$$a = \frac{\Delta v}{\Delta t} = \frac{v}{r}\frac{\Delta \ell}{\Delta t}$$

But $\frac{\Delta\ell}{\Delta t}$ is precisely the speed of the particle as it traverses the arc $\Delta\ell$, so we have derived that the magnitude of the centripetal acceleration is

$$\boxed{a_c = \frac{v^2}{r}} \tag{8}$$

We emphasize this result with a box: In *any* motion in which an object follows the arc of a circle, there *must* be an acceleration directed into the center of the circle with magnitude given by (8). After all, if there were no acceleration, then the object is required to continue with constant velocity, so it must keep going in the same direction.

Note that in circular motion the velocity is always tangent to the circle whereas the acceleration is always orthogonal to that. This means that the displacement is orthogonal to the centripetal acceleration. We will return to this point below when we consider the forces responsible for this centripetal acceleration.

So far, we have considered circular motion in which the speed of the particle was constant. This "uniform" circular motion has a well-defined *period* of time T it takes to make one revolution. The reciprocal $f = \frac{1}{T}$ of this period is called the *frequency*; its units are reciprocal seconds, also known as "Hertz".

When the frequency is expressed in terms of radians per second, it is called the *angular frequency* denoted ω (omega):

$$\omega = \frac{2\pi}{T} = 2\pi f$$

In terms of angular motion, this variable is the time rate of change of the angle, so another way of writing it is

$$\omega = \frac{\Delta\theta}{\Delta t}$$

This form is more general since it is valid also for motion which does not consist of a single revolution.

Exercise 43. *Show that the angular velocity ω for a particle in uniform circular motion of radius r and speed v satisfies*

$$\omega = \frac{v}{r} \qquad \Leftrightarrow \qquad \boxed{v = r\omega}$$

Rolling without slipping Consider now the motion of a wheel of radius r rolling uniformly along a level surface without sliding. This is an example of uniform circular motion. The time it takes the wheel to make one revolution is the period T. Its angular speed is $\omega = \frac{2\pi}{T}$, so a point on the edge of the wheel is moving relative to the fixed axis at a speed $v = r\omega$.

Since the wheel is not slipping, this point must have zero relative velocity with respect to the ground. This means the axel must have speed

$$\boxed{v = r\omega}$$

relative to the ground. In other words, a wheel that is rolling without slipping has translational velocity relative to the ground with magnitude $v = r\omega$.

In this story, it is important that the motion of the particles on the edge of the wheel is circular motion: This is what introduces the geometry relating the translation to the rotation. However, at no point was it relevant that the motion was *uniform*. A wheel that is rolling without slipping can accelerate provided the translational acceleration is accompanied by an angular acceleration that guarantees the wheel is not slipping.

Exercise 44. *Define*

$$\alpha = \frac{\Delta\omega}{\Delta t}$$

to be the angular acceleration of an object in circular motion. Show that when applied to the wheel above, the translational and angular acceleration are related by $a = r\alpha$.

Non-uniform circular motion In the previous exercise, we generalized to the situation in which the motion is circular, but the angular velocity is not required to be constant. Analogously to the acceleration, we defined the *angular acceleration* to be the rate of change of the angular velocity. Since the motion is circular, the velocity vector of the particle will change direction just as it had for uniform circular motion, but its length will change in time. Define the tangential acceleration

$$a_t = \frac{\Delta v}{\Delta t}$$

to be the rate of change of the *speed* of the particle in circular motion.

Exercise 45. *Describe the motion of a particle in non-uniform circular motion in detail. In particular, describe the direction of the acceleration and show that the tangential component has magnitude*

$$a_t = r\alpha$$

Exercise 46. *A wheel of radius* $r = .90\ m$ *is rolling without slipping along a level road subject to friction and stopping after a distance* $d = 155\ m$. *Its initial angular velocity is* $\omega = 6.2\ \frac{rev}{s}$. *What is the linear acceleration of the wheel?*

The *total acceleration* is the vector sum of the centripetal and tangential component vectors. By geometry, these are always perpendicular to one another. Therefore, circular motion that is not necessarily uniform, the magnitude of the acceleration is given by

$$a = \sqrt{a_c^2 + a_t^2}$$

Example 9. *A coin sits on a static turntable at a distance* $r = 9\ cm$ *from the center. The coefficient of static friction is* $\mu_s = 0.67$. *The turntable is switched on accelerating at an angular acceleration of* $\alpha = 1.23\ \frac{rad}{s^2}$. *At what time* t *does the coin begin to slide?*

Solution. *The angular velocity is $\omega = \alpha t$, so the centripetal component of the acceleration is $a_c = r\omega^2 = r\alpha^2 t^2$ whereas the translational one is $a_t = r\alpha$. The total acceleration has magnitude-squared $a^2 = a_c^2 + a_t^2 = r^2\alpha^4 t^4 + r^2\alpha^2$. The maximum force that static friction can apply is $\mu_s mg$ which will be equal to ma. Thus $\mu_s g = a$ or*

$$\mu_s^2 g^2 = r^2\alpha^4 t^4 + r^2\alpha^2 \qquad \Rightarrow \qquad t^4 = \frac{\mu_s^2 g^2 - r^2\alpha^2}{r^2\alpha^4} \approx 22.6s^4$$

or, taking the fourth root, $t \approx 2.2$ s.

Example 10. *A car accelerates uniformly from rest around a circular road with radius r.*

1. *At what angle from its starting point does the centripetal component of its acceleration equal the radial component?*
2. *At what time does this happen?*

Solution. *Starting from rest, $\omega_i = 0$ so a_c is initially 0 so its acceleration is entirely tangential. Since it is accelerating uniformly, its this component is $a_t = r\alpha$. Over time, the angular speed is increasing according to $\omega = \alpha t$ so the centripetal component grows. As a function of angle $\omega^2 = 2\alpha\Delta\theta$, so the centripetal component of the acceleration is growing as $a_c = r\omega^2 = 2r\alpha\Delta\theta$. The ratio*

$$\frac{a_c}{a_t} = \frac{2r\alpha\Delta\theta}{r\alpha} = 2\Delta\theta$$

Therefore, the centripetal component begins to exceed the tangential one already at $\Delta\theta = \frac{1}{2}$ radians, or $\Delta\theta \approx 29°$. In terms of time, $\frac{a_c}{a_t} = \frac{r\alpha^2 t^2}{r\alpha} = \alpha t^2$, so this corresponds to at time $t = \frac{1}{\sqrt{\alpha}}$.

You will notice from this example, that in most cases, a_t is negligible compared to a_c. When that is true, then $a = \sqrt{a_c^2 + a_r^2} \approx a_c$. In terms of answering questions about the acceleration on tests, be weary of any motion with relatively small angles (like 29°), short times, and/or lots of significant figures.

5.2 Roadmap

In our investigations of translational motion, we introduced the notion of linear momentum and its conservation in isolated systems. Non-conservation was due to net forces acting on non-isolated systems as codified in the Impulse-Momentum Theorem. We were also led to formulate the analogous notions of net work and kinetic energy as related in the Work-Kinetic Energy Theorem.

1. kinematic variables x, t, v, a
2. constant acceleration kinematic equations 1, 2, 3

	position	velocity	acceleration	inertia	momentum	force
Linear	x	$v = \frac{\Delta x}{\Delta t}$	$a = \frac{\Delta v}{\Delta t}$	m	$\boldsymbol{p}$	$\boldsymbol{F}$
Angular	θ	$\omega = \frac{\Delta \theta}{\Delta t}$	$\alpha = \frac{\Delta \omega}{\Delta t}$	I	$\mathbf{L}$	$\boldsymbol{\tau}$
Name	angle	angular velocity	angular acceleration	rotational inertia	angular momentum	torque

Table 1: Dynamical variables for linear and rotational motion

3. inertial mass and linear momentum
4. force
5. kinetic energy

In this section, we will repeat this process for rotational motion. Just as translational motion is formalized in the context of inertial frames as a shift of the origin of one such frame to another, rotations arise due to the possibility of rotating axes around the origin instead.

1. kinematic variables θ, t, ω, α
2. constant acceleration kinematic equations 1, 2, 3
3. rotational inertial and angular momentum
4. torque
5. rotational kinetic energy

5.2.1 Rotational Kinematics

When we rotate something, we are always doing it "around a point and "in a plane". Turning your notebook so that the person across your workbench from you can read it is a 180° rotation of your notebook in the plane of the workbench.

If we coordinatize the workbench as the xy-plane, we can introduce polar coordinates and describe the rotation in terms of an angle θ. By convention, we will choose the counter-clockwise

direction to be positive and we often pick $\theta = 0$ to be the $+x$-direction. Such a plane with a sense of direction is called an oriented plane.

The right-hand rule It turns out that working with oriented planes is often somewhat cumbersome, but in three spacial dimensions, there is a trick that can be used to assign a vector to every such plane and *vice versa*.

Consider again the bench with a rotation in that plane around a point that is the origin of a polar coordinate system. In three dimensions, there is a unique line orthogonal to the plane of the bench that passes through the origin. This shows that to every rotation, we can uniquely assign a line. To assign a vector, we need to pick a direction on this line either into the plane of the bench or out of the plane of the bench. The convention for a counter-clockwise rotation is to pick the direction pointing out of the bench.

This direction agrees with the following ritual: Using your open right hand, curl your fingers so that they describe the rotation. In this case, you can think of putting your right fist on the bench with your curled fingers rotating counter-clockwise. If you now extend your thumb, it will be pointing out of the bench in the same direction as the vector we just chose to describe the rotation.[15]

You can do the opposite as well: If you have a vector, say the displacement vector from your position to the exit door of the room, you can take your right hand, make a fist and line the direction of your thumb up with that vector. Then the base of your fist is flush with a plane orthogonal to that vector and oriented in the direction of your curled up fingers. As seen from the door, the plane is oriented counter-clockwise, as before.

Now consider two vectors $\boldsymbol{u}$ and $\boldsymbol{v}$. If they are parallel, they describe a line and there must be some scalar α such that $\boldsymbol{u} = \alpha\boldsymbol{v}$. If they are not parallel, then there is a plane in which they both lie. Putting $\boldsymbol{u}$ and $\boldsymbol{v}$ with their tails at the same point, we can apply the right-hand rule (RHR) to find a new vector describing the plane: Take your right hand and extend your fingers in the direction of the first vector $\boldsymbol{u}$. Then rotate your hand to whatever position you need it to be in so that when you curl your fingers, they point toward the direction that the second vector $\boldsymbol{v}$ is pointing. (This takes some practice; the instinct is to try to force the vectors to your hand's position, but you much move yourself so that you are oriented however the vectors dictate instead.) Once you have your (right!) hand in the correct position, your fist will be describing the oriented plane and your thumb will be pointing in the direction associated with that orientation.

Example 11. *Suppose $\boldsymbol{u}$ is the displacement vector from you to the exit door and $\boldsymbol{v}$ is the displacement*

[15] If you had accidentally used your left hand, your fist would originally have had to be "up-side down" and your thumb would end up pointing into the bench instead. This is a perfectly acceptable convention as well, but it is not the one chosen historically.

vector from you to the teacher's desk. Then the vector defined by the right-hand rule points directly upward from where you are sitting.

Definition 10. Let $\boldsymbol{u}$ and $\boldsymbol{v}$ be 3-dimensional vectors as before. We define the *cross product* $\boldsymbol{u} \times \boldsymbol{v}$ to be that vector pointing in the direction defined by applying the RHR to $\boldsymbol{u}$ followed by $\boldsymbol{v}$ (in that order) with magnitude given by $||\boldsymbol{u} \times \boldsymbol{v}|| = uv \sin\theta$ where θ is the angle between the vectors $\boldsymbol{u}$ and $\boldsymbol{v}$ in the plane spanned by them.

5.3 Energy

5.3.1 Rotational inertia from kinetic energy

Consider again the particle in circular motion with radius r and angular velocity ω. The tangential velocity of the particle is $v = r\omega$. Recall that when we studied kinetic energy, it was not important whether the motion was uniform or not, circular or otherwise. Thus, the kinetic energy of this particle is, as it always has been $K = \frac{1}{2}mv^2$. According to our analogy roadmap table 1, the analog of inertial mass is rotational inertia I. The analog of linear velocity is angular velocity, so the rotational analog of kinetic energy should be

$$K = \frac{1}{2}I\omega^2.$$

Kinetic energy is kinetic energy irrespective of the motion, so this must agree with our previous expression. Expanding out $K = \frac{1}{2}mv^2 = \frac{1}{2}m(r\omega)^2 = \frac{1}{2}(mr^2)\omega^2$, we find that the rotational inertia of a particle in circular motion is

$$I_{particle} = mr^2$$

If we have a bunch of particles, say N of them, with masses m_i all rotating at constant distances r_i from a common axis of rotation, then their inertia all add to give

$$I = \sum_{i=1}^{N} m_i r_i^2 \tag{9}$$

For example a hoop of total mass M is just a bunch of particles with masses m adding up to $M = \sum m$ all at a distance R away from the center and that therefore

$$I_{hoop} = MR^2$$

as well. Furthermore, we argued that by dimensional analysis any other distribution of mass M defined by a maximum length R had to have the same MR^2 form but with a coefficient that was smaller than 1. This is because any mass not at the maximum distance will have $r < R$ so these

terms contribute less to the sum (9). It is convenient to summarize this by saying that there must be some ε for which

$$I_\varepsilon = \varepsilon M R^2$$

Some examples are:

particle/hoop A mass M distributed entirely around the constant radius R has $\varepsilon = 1$.

disk/cylinder A mass M distributed as a uniform solid disk or cylinder of radius R has $\varepsilon = \frac{1}{2}$. Note that it does not matter how thick/tall the disk/cylinder is.

ball A (solid) ball of mass M and radius R (like a bowling ball or marble) has $\varepsilon = \frac{2}{5}$.

sphere A (hollow) sphere of mass M and radius R (like a ping-pong ball or basket ball) has $\varepsilon = \frac{2}{3}$.

rod A rod of length R and mass M has $\varepsilon = \frac{1}{12}$ when rotated through its center but $\varepsilon = \frac{1}{3}$ when rotated through one of its two ends.

Note that in the last entry, it matters around which axis we are trying to spin the object. Actually, this is true for any of the shapes: All the rotational inertiæ given above are specific to a natural axis. You need more information to calculate the rotational inertial around another axis. We will return to this point below when we introduce the Parallel Axis Theorem.

Example 12. *A round object with rotational inertial I rolls from rest down an incline without slipping. If the vertical distance it traveled is h, how fast is it going when it gets to the bottom?*[16]

Solution. *This question looks impossible to answer since you are only told the height and the moment of inertia. Let's just introduce whatever else we need and see how far we get with energy methods. Introducing the mass m of the object, the change in gravitational potential energy in this process is mgh. This work results in a final kinetic energy, which on general grounds is $K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$. Since the object is round, it should have some radius we will call r. Since it rolls without slipping, its speed will be $v = r\omega$. Finally, by dimensional analysis, there should be some number ε such that $I = \varepsilon m r^2$. This gives for the kinetic energy $K = \frac{1}{2}mv^2 + \frac{1}{2}\varepsilon m r^2\omega^2 = (1+\varepsilon)\frac{1}{2}mv^2$. Putting it together with the work done by gravity (Work-KE Theorem)*

$$mgh = (1+\varepsilon)\frac{1}{2}mv^2$$

Aha! The mass cancels out, as we are used to by now, but in addition the radius only appears through the combination $\varepsilon\frac{I}{mr^2}$. Therefore

$$v = \sqrt{\frac{2gh}{1+\varepsilon}} \tag{10}$$

[16] If such a problem is given in terms of angle θ and distance d along ramp, substitute $h = d\sin\theta$.

This is a remarkable result! Not only does the speed not depend on the mass, it doesn't even depend on the size! Besides the height, the only thing it depends on is how the mass is distributed within the object, not the size or the mass separately.

An immediate corollary of this analysis is that anything that can slide down the ramp without having to roll will have the highest speed at the bottom. For example, a cart acts like a frictionless object "sliding" down the ramp. (In reality, it has wheels that roll without slipping, but these are specifically designed to have very low inertia.) Let us also note that this relation is true for all heights y between 0 and h, so the faster object will have the higher speed throughout the motion and therefore will always be leading throughout the motion.

Therefore, in a race to the bottom between a cart, a hoop, a disk, a ball (solid), and a sphere (hollow), in what order to they arrive? Their order is by increasing ε: cart ($\varepsilon = 0$), ball ($\varepsilon = \frac{2}{5} = 0.4$), disk ($\varepsilon = \frac{1}{2} = 0.5$), sphere ($\varepsilon = \frac{2}{3} \approx 0.7$), hoop ($\varepsilon = 1$).

Exercise 47. *In a general motion with translation and rotation, what fraction of the total kinetic energy is due to rotation?*

5.3.2 Center of mass

The objects described above are rigid objects as consisting of constituent masses $m_1, m_2, ..., m_N$ located at positions $\boldsymbol{r}_1, \boldsymbol{r}_2, ..., \boldsymbol{r}_N$. Typically, N is a very large number since macroscopic objects consist of particles many times Avogadro's number. The total mass of the object is obtained by adding up all the constituent masses

$$M = \sum_{j=1}^{N} m_j = m_1 + m_2 + \cdots + m_N$$

In geometry, we sometimes talk about the geometric center of a shape. It is found by averaging the positions of the points. The language may be different, but the answer can be written as $\boldsymbol{r}_{center} = \frac{1}{N}\sum \boldsymbol{r}_j$. For example the center of a circle is exactly where you think it is because for every point $\boldsymbol{r}_j$ on the circle, there is another point on the opposite side with coordinate $-\boldsymbol{r}_j$. Adding them all up, you get pairwise cancelation, so $\boldsymbol{r}_{center} = \boldsymbol{0}$ (provided you center your coordinates at $\boldsymbol{0}$).

In a physical object, we also want to find the center, but it can happen that the physical object is not uniform. When that is the case the constituent mass m_j at a point $\boldsymbol{r}_j$ for some j is not equal to the mass m_i at some different point $\boldsymbol{r}_i$ with $i \neq j$. Because of this, the effect that forces have on this object will be different at different points. To find the overall point where the average forces act, need to find the *center of mass*. We do this not by taking the sum of the constituent coordinates $\{\boldsymbol{r}_j\}_{j=1}^{N}$ but the *weighted* sum

$$\sum_{j=1}^{N} m_j \boldsymbol{r}_j$$

This gives an average vector, but it does not have the units of length, so we divide by the total mass $M = \sum_{j=1}^{N}$ and consider the following quantity

$$\boldsymbol{r}_{CM} = \frac{1}{M}\sum_{j=1}^{N} m_j \boldsymbol{r}_j$$

Note that for a uniform object, all the m_j are equal $m_j = \frac{M}{N}$ for all $j = 1, ..., N$, and the center of mass reduces to the geometric center of the object.

We can now define the velocity of the CM and the acceleration of the CM. Assuming the constituent masses are not changing in time, these are just what you would expect:

$$\boldsymbol{v}_{CM} = \frac{1}{M}\sum_{j=1}^{N} m_j \boldsymbol{v}_j$$

and

$$\boldsymbol{a}_{CM} = \frac{1}{M}\sum_{j=1}^{N} m_j \boldsymbol{a}_j$$

Note that the sum of the momenta of a system of particles is equal to the total mass of the system times the CM velocity

$$\sum_{j=1}^{N} m_j \boldsymbol{v}_j = M\boldsymbol{v}_{CM} = \boldsymbol{p}_{CM}$$

Exercise 48. *Prove Newton's Second Law for the center of mass of a composite object: The net force* $\boldsymbol{F}_{net} = \sum_j \boldsymbol{F}_j$ *on a system of particles is given by the total mass of the system times the acceleration of the center of mass*

$$\boldsymbol{F}_{net} = \frac{\Delta \boldsymbol{p}_{CM}}{\Delta t} = M\boldsymbol{a}_{CM}$$

We can draw some powerful conclusions from these results: The importance of this statement is that, no matter how complicated the shape of a composite object, its center of mass satisfies Newton's Laws.

Example: Center of mass of a rod We will now calculate the center of mass of a rod of mass M and length L by chopping it into N pieces of mass $m = \frac{M}{N}$ and lengths $\ell = \frac{L}{N}$. Putting the 0 of coordinates on one end, the center of the jth piece is at $x_j = \frac{\ell}{2}(2j-1)$, so the center of mass of the rod is at

$$x_{CM} = \frac{1}{M}\sum_{j=1}^{N} m_j x_j = \frac{m}{M}\frac{\ell}{2}\sum_{j=1}^{N}(2j-1) = \frac{\ell}{2N}\left(2\sum_{j=1}^{N} j - \sum_{j=1}^{N} 1\right) = \frac{\ell}{N}\sum_{j=1}^{N} j - \frac{\ell}{2}$$

In the last step, we did the sum $\sum_{j=1}^{N} 1 = 1 + 1 + \cdots 1 = N$. To do the other sum, there is a trick: First take *double* the sum and then rearrange it like this:

$$\begin{aligned} 2\sum_{j=1}^{N} j &= 2(1+2+3+\cdots+N) = [1+2+3+\cdots+N] + [N+(N-1)+(N-2)+\cdots+1] \\ &= (N+1)+(N+1)+\cdots+(N+1) \qquad (N \text{ times}) \\ &= N(N+1) \end{aligned}$$

Therefore, we have demonstrated the useful result that adding up N consecutive integers gives

$$\sum_{j=1}^{N} j = \frac{N(N+1)}{2}$$

Returning to our calculation of the center of mass, we find

$$x_{CM} = \frac{\ell}{N}\frac{N(N+1)}{2} - \frac{\ell}{2} = \frac{\ell}{2}(N+1) - \frac{\ell}{2} = \frac{\ell N}{2} = \frac{L}{2}$$

Of course this is not only a silly calculation to do, it is a silly way to do it. Had we put the origin of the x coordinates in the middle, the mass m_j at x_j would give a contribution that is canceled by a term on the other side with coordinate $-x_j$. This would tell us that $x_{CM} = 0$, which is the same point as $x_{CM} = \frac{L}{2}$ in the previous system of coordinates. The point here is just to see how an actual calculation would go so that we might have a better understanding of related results.

Example (cont'd): Moment of inertia of a rod Recall that the rotational inertia is defined by

$$I = \sum_{j=1}^{N} m_j (r_\perp)_j^2$$

where $r_\perp$ is the distance to the axis of rotation. For the rod around an end, this is $I = \sum_{j=1}^{N} m_j x_j^2$. We can repeat the type of calculation above for this quantity:

$$I = \sum_{j=1}^{N} m_j x_j^2 = \frac{M}{N}\frac{\ell^2}{4}\sum_{j=1}^{N}(2j-1)^2 = \frac{M}{N}\frac{\ell^2}{4}\sum_{j=1}^{N}(4j^2-4j+1)$$

We now have three sums to do. Two of them, we know how to do, but the third one $\sum_{j=1}^{N} j^2$ is new. We will not continue with this calculation, but any of you who would like a challenge are invited to look up square pyramidal number. Using that result, you would be able to demonstrate that, for large N, the moment of inertia approaches

$$I_{rod,end} = \frac{1}{3}ML^2 \tag{11}$$

as the number of parts $N \to \infty$ gets large.

Exercise 49. *An annular cylinder is a cylindrical with a cylindrical hole down the middle along the axis of symmetry. It is a "fat" hoop, in which the thickness of the hoop cannot be ignored. In the table listing the rotational inertias of various round shapes, the inertia of the annular cylinder is listed as* $I = \frac{1}{2}m(r_1^2 + r_2^2)$ *where* m *is the mass,* r_1 *is the "inner" radius (the radius of the hole) and* r_2 *is the "outer" radius.*

1. *Take limits* $r_1 \to 0$ *and* $r_1 \to r_2$ *to confirm that you recover the correct rotational inertiae for the resulting shapes.*
2. *We can think of the annular cylinder as a large cylinder with a smaller inner cylinder removed from it. Derive the formula for the rotational inertia of the annular cylinder as follows:*
   (a) *Calculate the volume* $V_\mathbf{o}$ *of the annular cylinder. Denote the height by* ℓ.
   (b) *Define the mass density* $\rho = \frac{m}{V_\mathbf{o}}$ *. This is the mass per unit volume.*
   (c) *Calculate the rotational inertia of a cylinder of radius* r_2*, height* ℓ*, and density* ρ.
   (d) *Subtract from this the rotational inertia of a smaller cylinder of radius* r_1*, height* ℓ*, and density* ρ.
   (e) *Show that your expression is equivalent to* $I = \frac{1}{2}m(r_1^2 + r_2^2)$.

   Hint: For this part of the problem, it may be easier to start with some explicit numbers like $r_1 = 10\,cm$*,* $r_2 = 20\,cm$*,* $\ell = 30\,cm$*, and* $m = 7\,kg$*, and go through the steps before attempting the same feat with variables.*

5.3.3 Parallel Axis Theorem

As we have seen, the rotational inertia–and therefore, the moment of inertia–depends on the axis around which we rotate. Suppose that we choose the axis to pass through the center of mass and that the moment of inertia through this axis is I_{CM}. The parallel axis theorem states that the moment of inertia I through a different axis *parallel* to the first is given by $I = I_{CM} + Md^2$, where d is the distance between the original axis (through the center of mass) and the new one. This is the

Theorem 8 (Parallel Axis Theorem)**.** *Let* I_{CM} *denote the moment of inertia of an object of mass* M *through an axis passing through the center of mass of the object. Let* I *denote the moment of inertia around an axis parallel to the first and separated from it by a distance* r*. Then*

$$I = I_{CM} + Mr^2 \tag{12}$$

Exercise 50. *Use* (11) *and the Parallel Axis Theorem to show that*

$$I_{rod,middle} = \frac{1}{12}ML^2 \tag{13}$$

for a rod mass M *and length* L *rotated through its center.*

We can confirm this by modifying our explicit calculation for I, but let us be lazy and make the following clever argument: The sum for I_{CM} is modified because $r_\perp$ is now the distance to the center of mass $x_{CM} = L/2$ instead of the distance to one end. We can still use our sum if we recognize that we can start at $x = 0$ and sum only the first $N/2$ terms to get to the middle, and finally multiply the whole thing by 2 to get both sides:[17]

$$I = 2\sum_{j=1}^{N/2} m_j x_j^2$$

In the explicit calculation we did previously, w saw that the important part of the sum scales like $\sum_{j=1}^{N} j^2 \sim \frac{1}{6}N^3 + ...$, so if we replace $N \to N/2$, we get an extra factor of $1/2^3 = 1/8$. However, we doubled the sum, so the relevant factor is reduced to $1/4$. Because of this, $I_{CM} \to \frac{1}{4}I = \frac{1}{12}ML^2$. All the terms we ignored have additional factors of $1/N$, so they all vanish when we take the number of parts $N \to \infty$. Thus, we can see more-or-less explicitly where the extra factor of $1/4$ comes from, and we have confirmed the result (13).

Exercise 51. *Prove the following corollary of the Parallel Axis Theorem:*

Corollary 2. *When the moment of inertia is minimal, the rotation axis passes through the center of mass.*

Let us end by remarking on the structure of the Parallel Axis Theorem: The equation for I reduces to I_{CM} when $r \to 0$, as required. Furthermore, if the object in question is a point mass or a linear mass lying along the axis of rotation, it has no intrinsic moment of inertia so $I_{CM} \to 0$ and $I \to Mr^2$ which is the correct moment of inertia for a mass M in rotation with radius r. Thus, we can think of the Parallel Axis Theorem as stating that the inertia of an object is a combination of it's "intrinsic" inertia I_{CM} and the rotational inertia of its orbital motion.

Example 13. *In class, we discussed the kinetic energy of a wheel $K = K_{trans} + K_{rot}$. Here $K_{trans} = \frac{1}{2}mv^2$ is the translational part and $K_{rot} = \frac{1}{2}I\omega^2$ is the rotational part. If the wheel is modeled by a hoop, then $I = mr^2$ where m and r are the mass and the radius, respectively. If the wheel is rolling without slipping, we argued that then $v = r\omega$. Therefore, the rotational inertia $K_{rot} = \frac{1}{2}mr^2\omega^2 = \frac{1}{2}mv^2 = K_{trans}$, so the total kinetic energy is $K = 2K_{trans}$ in this case.*

We can show this a different way by using the parallel axis theorem to calculate the rotational inertia of the wheel around the stationary point touching the ground. Since that point is a distance r removed from the axis of the wheel, we get that the rotational inertia is shifted to

$$I = I_{CM} + mr^2 = 2mr^2.$$

[17] If N is not even, we can either split off the first term in explicitly or sum only to $\frac{1}{2}(N-1)$. Either way, we will get a correction that goes $\to 0$ as $N \to \infty$.

Then the rotational kinetic energy due just to this type of "rotation" is $K = \frac{1}{2}2mr^2\omega^2 = 2\frac{1}{2}mv^2 = 2K_{trans}$.

Exercise 52. *Repeat the arguments above for the case of a cylinder/solid disk, a solid ball, and a hollow sphere.*

Exercise 53. *Prove the Parallel Axis Theorem in 1D by taking the following steps: Suppose* I_{CM} *is known. In general* $I_{CM} = \sum_i m_i r_i^2$*, so although we do not know each term in the sum, we are saying we do know the result* I_{CM} *is a specific amount.*

- *Choose your coordinates* r_i *so that the origin is the center of mass, that is* $r_{CM} = 0$*. Write down the equation for the center of mass in these coordinates.*
- *Shift from the center of mass by adding a fixed vector* d *to all your coordinates. Argue that the rotational inertia I around this new point is gotten from the formula for* I_{CM} *by shifting* $r_i \to r_i + d$.
- *Expand out the factors for* $(r_i + \mathrm{d})^2$ *in the formula for I into three terms. Notice that your formula looks like* $I = I_{CM} + M\mathrm{d}^2$*+something linear in* d.
- *Stare at your term that is linear in* d *until you recognize why it vanishes.*

Once you have carried out these steps, you have demonstrated that the parallel axis theorem holds in 1D. The full 3D version is exactly the same argument but involving uglier algebra.

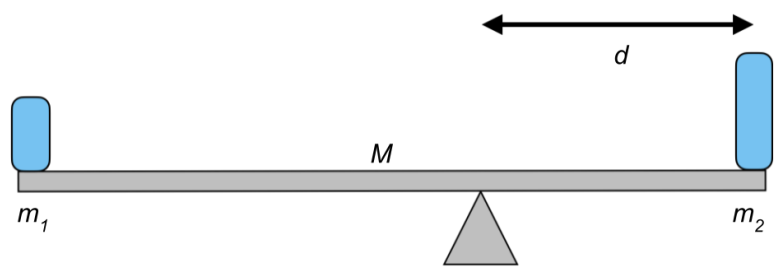


Figure 22: Beam with masses

Example 14. *Figure 22 shows a configuration in which a beam of mass M and length* ℓ *has two masses* m_1 *and* m_2 *placed at its ends. The fulcrum is a distance* d *from the right side. Calculate the rotational inertia of this system.*

Solution. *The trap lurking here is that we cannot forget that the beam has a rotational inertia* $I_{CM} = \frac{1}{12}M\ell^2$ *of its own, and that this is changed by the fact that the fulcrum lies a distance* $\frac{\ell}{2} - \mathrm{d}$ *from the center of mass. Thus, the parallel axis theorem implies that* $I_{fulcrum} = \frac{1}{12}M\ell^2 + M\left(\frac{\ell}{2} - \mathrm{d}\right)^2$

just for the beam. In addition to this, we must add the rotational inertiæ of the two masses to obtain $I = I_{fulcrum} + m_1(L-d)^2 + m_2 d^2$. *Finally, then, we find for the rotational inertia of the whole system about the fulcrum*

$$I = \frac{1}{12}M\ell^2 + M\left(\frac{\ell}{2} - d\right)^2 + m_1(\ell - d)^2 + m_2 d^2$$

5.4 Angular Momentum

The next concept after inertial mass to be defined in our progression through linear dynamics was linear momentum $\boldsymbol{p}$, Newton's concept of "quantity of motion". When the object had a constant mass m, the momentum could be written in terms of it and the velocity as $\boldsymbol{p} = m\boldsymbol{v}$.

Continuing with our analogy, the rotational analog of speed is angular speed ω, and we have just concluded that the rotational analog of inertial mass is rotational inertia I. We, therefore, propose to call the rotational analog of linear momentum *angular momentum* L and to define it as

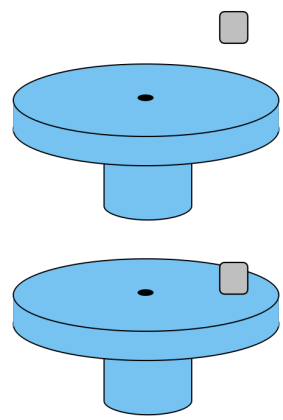


Figure 23: A lump of clay of mass m is dropped on a spinning plate with rotational inertia I in example 16.

Definition 11 (Angular momentum)**.** The angular momentum L of an object of rotational inertia I and angular velocity ω is

$$L = I\omega$$

In class, we spun around on the heavy rotating plate with 1 kg masses in each hand. We held our arms out, initiated the spin, and then pulled our arms in. When we did that, we were reducing our rotational inertia, and we noticed that our rotational speed increased. We conjectured that in the process, the product $I\omega$ stays constant, and concluded that this must be the rotational analog of conservation of linear momentum:

For an isolated system, angular momentum is conserved.

This is the rotational analog of the law of conservation of linear momentum for an isolated system.

Example 15. *Suppose when your arms are pulled in you have a rotational inertia of* $I_1 = 20\ kg \cdot m^2$, *but when they are outstretched the inertia increases to* $I_2 = 25\ kg \cdot m^2$. *Suppose that with outstretched arms, you are rotating on the spinning platter with angular speed* $\omega_2 = 0.5\ Hz$. *Calculate the work you do when you pull in your arms.*

Solution. *By conservation of angular momentum* $I_1\omega_1 = I_2\omega_2$ *so* $\omega_1 = \frac{I_2}{I_1}\omega_2$. *By the work-kinetic energy theorem, the work you do is* $W_{net} = \frac{1}{2}I_1\omega_1^2 - \frac{1}{2}I_2\omega_2^2$. *So*

$$W_{net} = \frac{1}{2}\frac{I_2}{I_1}(I_2 - I_1)\,\omega_2^2$$

gives $W_{net} \approx 0.78$ *J.*

Example 16. *In this problem we have a spinning pottery wheel with rotational inertia I and angular velocity* ω_i. *On it, we drop from rest a lump of clay of mass* m. *It hits the wheel and sticks to it at a radius* r *from the center of the wheel as shown in figure 23. What is the angular velocity of the plate with the lump of clay attached?*

Solution. *Before the clay is released, the angular momentum of the system is* $L_i =$. *After the clay is attached, the rotational inertia of the system is* $I + mr^2$ *and the final angular momentum is* $L_f = (I + mr^2)\omega_f$. *Dropping the clay onto the plate does not involve any external torque, so the system is isolated and conservation of angular momentum applies*

$$L_f = L_i \qquad \Rightarrow \qquad \omega_f = \frac{I}{I + mr^2}\omega_i$$

Note that $\frac{I}{I+mr^2} < 1$, *so the plate spins at a lower angular velocity after the drop.*

5.5 Torque

In the chapter on work and energy, we studied a problem in which the work done on a rock by a lever was equal to the work done by a force pushing on the other side of the lever.

In this example, the work due to the force applied is $W_{in} = \boldsymbol{F}_{in} \cdot \Delta\mathbf{r}_{in}$. Here the dot product takes only that part of the force along the displacement $\Delta\mathbf{r}_{in}$, or–equivalently–that part of the displacement along the force. If the part of the lever on the left has length ℓ_{in}, this gives $W_{in} = F_{in}\ell_{in}\cos\theta$. Repeating this calculation for the output on the right, we find $W_{out} = F_{out}\ell_{out}\cos\theta$. Setting these equal, we found that

$$F_{out} = \frac{\ell_{in}}{\ell_{out}}F_{in}$$

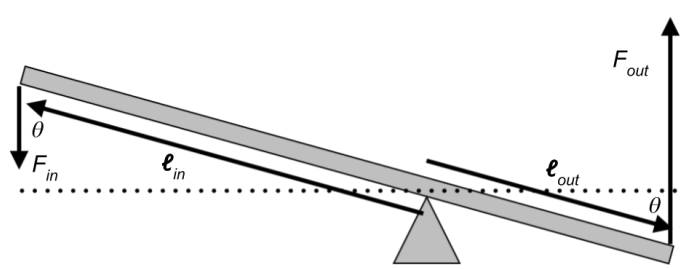


Figure 24: A lever as a force multiplier

and we concluded that as long as the lever on the left is longer than that on the right, we could get a larger force out than we put in.

Notice that this type of force causes a rotation about the pivot point (also called the "fulcrum"). Also notice, that if we take $\boldsymbol{\ell}_{in}$ to denote the vector pointing from the fulcrum to the point where the force is applied then

$$||\boldsymbol{\ell}_{in} \times \boldsymbol{F}_{in}|| = F_{in}\ell_{in} \sin\theta$$

The cross product computes the part of the force *orthogonal* to the pivot arm, or–equivalently–the length of the part of the arm that is orthogonal to the force. The other component of the force is directed toward the fulcrum and does not help turn the lever.

This type of turning force is called "torque". Notice that by construction is has the same units as work, but that is where the analogy ends. Where the work is a scalar quantity, the torque is a vector. In the example just described, that vector points in the direction out of the page. Since we want to emphasize that torque is different from work, we never use the units of energy to describe it. Instead, the units of torque are Newton-meters or foot-pounds or something equivalent. The Greek letter τ we will use for torque is pronounced "tau".

Definition 12 (Torque)**.** The torque $\boldsymbol{\tau}$ that a force $\boldsymbol{F}$ exerts on an object around a point is given in terms of the displacement vector $\mathbf{r}$ from that point to where the force is applied by

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$$

Note that this implies that the direction is given by the right-hand rule and the magnitude $\tau = rF\sin\theta$, where θ *denotes the angle between* $\mathbf{r}$ *and* $\boldsymbol{F}$.

Exercise 54. *Show that the magnitude of the torque is the part* $F_\perp$ *of the force perpendicular to* $\mathbf{r}$ *times the length of* $\mathbf{r}$

$$\tau = F_\perp r$$

Draw the triangle geometry that demonstrates this. Show that the magnitude of the torque is also the product of the magnitude of the force and the perpendicular distance $r_\perp$ *from the axis of rotation to the line along which the force acts.*

$$\tau = Fr_\perp$$

Draw this interpretation in your diagram as well.

Definition 13. The perpendicular distance $r_\perp$ from the axis of rotation to the line along which the force acts is called the *lever arm*.

For a given force, the maximal torque will result when we maximize the lever arm. In practice, we often try to adjust the force to maximize the lever arm by

1. apply the force perpendicularly to $\mathbf{r}$, and

2. make the lever arm as large as possible.

For example, when you need to open a heavy door, you will generally not push on the door anywhere near the hinge. Instead, you will push perpendicularly to the door as far away from the axis of the hinge as is practical.

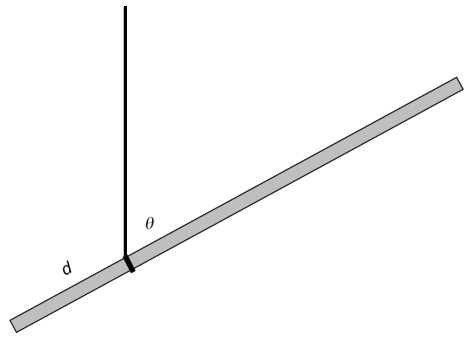


Figure 25: An unstable stick and string configuration

Example 17. *Consider a stick of length $\ell = 1.0\ m$ and mass $M = 86\ g$ attached to a string a distance $d = 30\ cm$ from one end and making an angle $\theta = 60°$ with it as in 25. If the stick is released from rest, what is the torque around the point where the string is attached?*

Solution. *Gravity acts with a force $F_g = Mg \approx 0.84\ N$ on the center of mass of the stick located at a distance $\frac{\ell}{2} \approx 50\ cm$ from the end of the end. The distance of this point from the pivot (where the string is attached) is $r = \frac{\ell}{2} - d \approx 0.20\ m$. The vector appearing in the torque is the one with this magnitude pointing from the pivot to the center of mass. The RHR tells us, then, that the direction the torque points is into the page. This is in line with our expectation that the stick will rotate clockwise.*

The angle between $\mathbf{r}$ *and* $\boldsymbol{F}_g$ *is* not θ. *Instead it is* $180° - \theta$. *However* $\sin(180° - \theta) = -\sin(-\theta) = \sin\theta$, *so we still find that* $\tau = rF\sin(180° - \theta) = rF\sin\theta \approx 0.15\ N \cdot m$.

Exercise 55. *A person is attempting to take the flat tire off of their car with the lug wrench provided in a compartment in the trunk for precisely this task. The lug nuts on the wheel are very tight and difficult to remove, so the person puts the wrench on and then stands on the end of it.*

1. *If the person has a mass of* $70\ kg$ *and the wrench has a length of* $40\ cm$, *what is the maximum torque (in standard units!) the person will be able to apply this way?*
2. *In what way is this person maximizing the lever arm?*
3. *Is there any way the person could apply more force?*
4. *What will happen if the person is successful and the bolt comes loose? Describe your answer using angular kinematical variables.*

Exercise 56. *You are pushing a person of mass $m = 25\ kg$ on the outer edge of a* massless *and frictionless merry-go-round of radius $r = 2\ m$ by applying a force $F = 100\ N$. Your goal is to gain maximum fun by generating maximum speed.*

1. *Describe in detail how you push, and draw a diagram.*
2. *Calculate the tangential acceleration of the person you are pushing.*
3. *Calculate the angular acceleration α of the merry-go-round.*
4. *Calculate the rotational inertia of the person on the massless merry-go-round.*
5. *Calculate the torque τ you are applying.*
6. *Show that $\tau = I\alpha$.*

This exercise does not describe a realistic scenario because the mass of a real merry-go-round is not 0. Suppose the mass of the merry-go-round is $M = 200\ kg$ and that its shape is that of a solid cylinder.

7. *Using what you learned above, try to recalculate the angular acceleration in this more realistic scenario.*

In this exercise, we found that a force acting on a particle of mass m at a fixed distance r from a central point will accelerate the particle according to $\tau = I\alpha$. For a rigid system consisting of an enormous number N of point particles, all that would change is that the moment of inertial would be the one due to all the particles $I \to \sum_{i=1}^{N} I_i$: Since the object is rigid, all points must have the same angular motion, so pushing one of them results in all of them having to accelerate at the same rate.

In general, there could be a bunch of forces acting on the merry-go-round each generating their own torque. Adding all the torques will result in a net torque $\boldsymbol{\tau}_{net} = \sum_i \boldsymbol{\tau}_i$. It is then this net torque that will cause an angular acceleration:

$$\boxed{\tau_{net} = I\alpha}$$

This is the rotational version of Newton's Second Law.

Example 18. *Consider again the stick and string from example 17. Calculate the angular acceleration when the stick is released.*

Solution. *This is an application of the parallel axis theorem: The torque $\tau_{net} \approx 0.15\ N \cdot m$ we calculated previously acts around the point held by the string a distance $d \approx 30\ cm$ from the end, or $r = \frac{L}{2} - d \approx 0.20\ m$ from the center of mass. The PAT then tells us that $I = I_{CM} + Mr^2 \approx 0.011\ kg{\cdot}m^2$ where $I_{CM} = \frac{1}{12}ML^2 \approx 7.2 \times 10^{-3}\ kg \cdot m^2$ was the value of the moment of inertia through the center of the stick. Using this value in Newton's second law for rotations, we calculate that the angular acceleration is $\alpha = \frac{\tau_{net}}{I} \approx 14\ rad/s^2$.*

Exercise 57. *You stop pushing the child on the merry-go-round in problem 56 when they have an angular velocity of* $33\ rpm$. *If it takes them a time of* $30\ s$ *before they come to rest, how large is the torque due to friction? If you were to start pushing again with* $100\ N$ *of force, how long (in seconds) would you be pushing to get the child back to* $45\ rpm$?

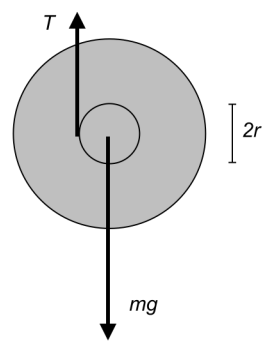


Figure 26: The inner workings of a yo-yo

Example 19. *A yo-yo is made of an inner cylinder with mass* $m_1 = 5\ g$ *and radius* $r_1 = 0.5\ cm$ *attached to two identical cylinders each with mass* $m_2 = 25\ g$ *and radius* $r_2 = 5\ cm$. *A string of length* $\ell = 1\ m$ *is wound around the inner cylinder and looped around a child's finger. If the child drops the yo-yo, how long does it take to de-spool completely? What is the tension in the string?*

Solution. *The inside of the yo-yo and the relevant forces are represented in figure 26. Applying Newton II, we find for the downward acceleration*

$$ma = mg - T$$

where $m = m_1 + 2m_2 = 0.055\ kg$ *is the mass of the yo-yo.*

Now we have a choice to make: We can either consider the rotational motion through the axis of symmetry of the yo-yo, or we can use the parallel axis theorem to consider pivot point to be the one where the tension is acting. For illustrative purposes, we will do both.

- *Around the axis of symmetry, the only force generating a torque is the tension because the moment arm for the weight is zero. (The weight vector passes through this point, so it generates no torque.) Thus*

  $$r_1 T = \tau_{net} = I\alpha$$

  where $I = \frac{1}{2}(m_1 r_1^2 + 2m_2 r_2^2) = 6.3 \times 10^{-5}\ kg \cdot m^2$ *is the rotational inertia through the center of the yo-yo. Together with the center of mass acceleration above, this gives* $ma = mg - \frac{I}{r_1}\alpha$.

The yo-yo rolls without slipping, so we get $ma = mg - \frac{I}{r_1^2}a$ *and, collecting* a *terms,*

$$a = \frac{1}{1 + \frac{I}{mr_1^2}} g \tag{14}$$

This gives a tension of (show this!)

$$T = \frac{I}{mr_1^2 + I} mg \tag{15}$$

- *We now redo this analysis but using the parallel axis theorem. Around the point where the tension acts, only the weight generates a torque. The parallel axis theorem tells us that around this point, the rotational inertia is* $I' = I + mr_1^2$*, so this time the angular equation is* $r_1 mg = (I + mr_1^2)\alpha$*. Again using rolling without slipping,* $\alpha = \frac{a}{r_1}$*, we get* $r_1 mg = (I + mr_1^2)\frac{a}{r_1}$*. Solving for* a *is easier now because all the* a*-terms are already collected, and we again find* (14).

Now that we have found the linear acceleration of the yo-yo, the rest is CAKE—CAKE 2 to be exact: $\ell = \frac{1}{2}a(\Delta t)^2 \Rightarrow \Delta t = \sqrt{\frac{2\ell}{a}}$

$$\Delta t = \sqrt{1 + \frac{I}{mr_1^2}}\sqrt{\frac{2\ell}{g}} \tag{16}$$

Our values for the mass and inertia imply $\frac{I}{mr_1^2} = 45.5$*, so in the end, we find* $\Delta t = 6.82 \times 0.45 = 3.1$ *s.*

We can simplify this since $m_2 = 5m_1$ *and* $r_2 = 10r_1$*. This gives* $I = \frac{1001}{2}m_1 r_1^2$ *and* $m = 11m_1$ *combining to give*

$$\Delta t = \sqrt{1 + \frac{1001}{22}}\sqrt{\frac{2\ell}{g}}$$

Besides simplifying the numerology,[18] *this shows that the masses and radii separately don't matter at all. In fact, if we take* $m_2 = \mu m_1$ *and* $r_2 = \rho r_1$*, then* $I = \frac{1}{2}(1 + 2\mu\rho^2)m_1 r_1^2$ *and* $m = (1 + 2\mu)m_1$*. This shows that the general formula*

$$\Delta t = \sqrt{1 + \frac{1 + 2\mu\rho^2}{2(1 + 2\mu)}}\sqrt{\frac{2\ell}{g}}$$

depends only on the ratios of the masses $\mu = \frac{m_2}{m_1}$ *and radii* $\rho = \frac{r_2}{r_1}$ *and not on each separately. In other words, it is the geometry of the yo-yo that matters and not its size or mass.*

[18] The first time I did this calculation, my angular acceleration was off by a factor of 10, giving a suspiciously small Δt. I found the error by working out the analytic form.

Note: Previously, we listed the values of $\varepsilon = \frac{I}{MR^2}$ for a series geometries with axial symmetry. In those cases, we always found that ε was bounded above by 1 (the hoop): How could it be larger than this if all masses are as far away from the axis of symmetry as possible? This example shows that it is possible for an object to have such a factor $\varepsilon > 1$ when the radius of the circle around which the object rolls is not the largest radius. In the yo-yo $r_1 < r_2$ and it is true that $I < Mr_2^2$, but since the radius at which the force is applied is r_1, the geometric factor relevant for the yo-yo is $\varepsilon = \frac{I}{Mr_1^2}$ which is larger than Mr_2^2 by a factor of $\left(\frac{r_2}{r_1}\right)^2$. In the example above, $\varepsilon = \frac{1001}{11} \approx 45.5$.

To be clear: The rotational inertia $I = 6.3 \times 10^{-3}\ kg \cdot m^2$ is just whatever it is measured to be. When we introduced ε, the logic was to compare this to the rotational inertia mr^2 where m is the mass of the object and r is whichever radius the object is rolling around without slipping (*i.e.* the one appearing in the formula $v = r\omega$). Although we cannot change I, some geometries allow us to roll around a circle with radius that is smaller than the overall size of the object. Precisely this is what makes the yo-yo a fun toy.

Exercise 58. *A cylinder of mass m and radius r has a long ribbon wrapped around it. The ribbon is attached to the ceiling with the cylinder held at rest directly beneath where the ribbon is attached to the ceiling. The cylinder is released from rest and begins to unspool as drops.*

1. *Find the time it takes to drop a distance of two meters. (Hint: If you have been paying attention, you can reuse a previous result.)*
2. *Find the angular acceleration of the cylinder as it falls, and use this to find the linear acceleration of the axis as it falls.*

Example 20. *Consider again the Atwood machine, but this time we will not assume the pulley is ideal. Instead, it will still be frictionless, but it will have a rotational inertia I. On the right, hangs a block of mass m_r and on the left hangs one with mass $m_l > m_r$.*

1. *Calculate the tension in the string on the left and the right sides.*
2. *Calculate the acceleration of the larger mass.*
3. *If the pulley has some friction with torque τ_f, what is the acceleration of the larger mass?*

Solution. *We begin by drawing a FBD. This time the pulley is included as a disk with forces acting at specific points. Newton's second law becomes*

$$m_l a = m_l g - T_l \qquad , \qquad I\alpha = \tau_{net} = (T_l - T_r)r \qquad , \qquad m_r a = T_r - m_r g$$

The structure of these equations is as follows:

- *The direction of positive acceleration is chosen as downward on the left, because that is the larger mass. The tension in the string is keeping the left mass from accelerating at* $9.8\ m/s^2$.

- *The acceleration of the mass on the right is upwards and the same size as the acceleration on the left because the string does not stretch. This is the opposite direction of gravity and along the right-side tension.*
- *The difference in tensions is causing a torque on the pulley. In terms of direction, the torque is out of the page, as determined by the RHR.*
- *Since the tensions are tangential, they perpendicular to the lines connecting the center to the points where the string is tangential. Thus, the moment arms are just r and the torque on the left is $T_l r$ out of the page while that on the right is $T_l r$ into the page. Thus $\tau_{net} = T_l r - T_l r$. This torque equals the rotational inertia of the pulley times the angular acceleration of the pulley by Newton's Second Law for rotations.*

We can solve for the acceleration by multiplying the first and third equations by r and then adding them all up.

$$(m_l + m_r)ra + I\alpha = (m_l - m_r)rg$$

Multiplying again by r to convert $\alpha \to a = r\alpha$ (rolling without slipping), we get

$$[(m_l + m_r)r^2 + I]a = (m_l - m_r)r^2 g \qquad \Rightarrow \qquad a = \frac{m_l - m_r}{m_l + m_r + \frac{I}{r^2}} g$$

Note that in the limit that the pulley is ideal ($I \to 0$), we recover (4). Just as in that case, the numerator is proportional to the difference in masses whereas the denominator is a corrected version of the total mass of the system: Recalling that $I = \varepsilon m r^2$ we see that $\frac{I}{r^2}$ is the mass of the pulley up to a geometric factor ε describing how the mass is distributed. (We expect ε to lie between $\frac{1}{2}$ for a solid disk and 1 for a wheel.)

The tensions now follow by plugging this into the first and third equations. The algebra is unpleasant, but the result can be made to look like

$$T_l = \left[\frac{2m_r + \frac{I}{r^2}}{m_l + m_r + \frac{I}{r^2}}\right] m_l g \qquad \text{and} \qquad T_r = \left[\frac{2m_l + \frac{I}{r^2}}{m_l + m_r + \frac{I}{r^2}}\right] m_r g$$

Again we see that in the limit $I \to 0$, these agree with the tension of the ideal Atwood machine (5).

Example 21. *If we drop a roll of toilet paper, it will accelerate toward the ground at $g = 9.8\ m/s^2$, but if we hold the end of the paper a fixed height above the ground and then drop the roll, it will accelerate less fast because it must unroll to fall. Suppose the roll has mass $164\ g$, an outer radius of $6.5\ cm$ and an inner radius of $2.0\ cm$. How long does it take the roll to fall a distance of $1.5\ m$? How much longer does it take than free fall?*

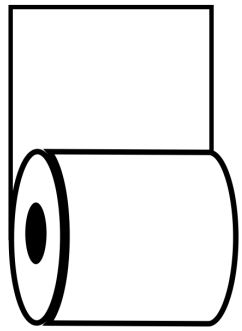


Figure 27: A roll of toilet paper.

Solution. *There are multiple ways to do this problem. One way that comes to mind is to use energy conservation to get the final speed and then to use that and CAKE to find the time. Another is to use torque and CAAKE to find the time. We will do both to illustrate the methods.*

To use energy, we can use either the conservation of mechanical energy or the work-KE theorem. The work that gravity does mgh *will be converted into kinetic energy* $K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ *where the first term is the translational KE and the second is the rotational part. Without the second part, we would just have the free-fall case.*

The TP rolls without slipping as it falls. Because of that, $v = r\omega$*. We can use this to combine the two kinetic terms* $K = \frac{1}{2}mv^2 + \frac{1}{2}\frac{I}{r^2}v^2 = \frac{m}{2}\left(1 + \frac{I}{mr^2}\right)v^2$*. Recall that* $\frac{I}{mr^2}$ *is the term we previously called* ε*. Setting this equal to the work done by gravity gives* $mgh = \frac{m}{2}(1+\varepsilon)v^2$*. Solving for* v*, we again get* (10)*!*

Should we be surprised? Well, in hindsight no, since the work done by gravity depends only on the height. What should definitely not be the same is the time that it takes to get to the floor, so let's see how that works. CAKE 3 gives us the acceleration as

$$a = \frac{v^2}{2h} = \frac{g}{1+\varepsilon}$$

(This is less than g*, unless* $\varepsilon = 0$*, so that makes sense.) Then CAKE 2 gives*

$$\Delta t = \sqrt{\frac{2h}{a}} = \sqrt{1+\varepsilon}\sqrt{\frac{2h}{g}} \tag{17}$$

Note that this is the same equation as (16)*. This time is larger than for free fall time* $\sqrt{\frac{2h}{g}} \approx 0.553\ s$ *by a factor of* $\sqrt{1+\varepsilon}$*.*

All that remains is to compute $\varepsilon = \frac{I}{mr^2}$*. In exercise 49, we derived the rotational inertia for this*

geometry $I = \frac{1}{2}m(r_1^2 + r_2^2)$. *This gives for*

$$\varepsilon = \frac{I}{mr_2^2} = \frac{1}{2}\left(1 + \frac{r_1^2}{r_2^2}\right) \approx 0.547$$

for a time of $\Delta t = \sqrt{1+\varepsilon}\sqrt{\frac{2h}{g}} \approx 0.69$ *s.*

Now, let us start over and use torque methods. First, we draw the FBD for this geometry which consists of a disk of radius r_2*, a weight vector vertically down from the center, and a tension upward and tangent to the disk.*

The tension will be less than the weight, so the roll will accelerate vertically downward according to Newton II. At the same time, the tension exerts a torque on the roll:

$$ma = mg - T \qquad \text{and} \qquad r_2 T = I\alpha$$

The second equation can be solved for $T = \frac{I}{r_2}\alpha$. *Since the TP rolls without slipping, we can rewrite this in terms of linear acceleration:* $T = \frac{I}{r_2^2}(r_2\alpha) = \frac{I}{r_2^2}a = \frac{I}{mr_2^2}ma = \varepsilon ma$. *Plugging this back into the first equation gives*

$$ma = mg - \varepsilon ma \qquad \Rightarrow \qquad a = \frac{g}{1+\varepsilon}$$

and we recover our previous solution. The remainder of the steps are the same as those above, but note that we have gained a new formula for the tension: Plugging this a *into* $T = \varepsilon ma$*, we find*

$$T = \frac{\varepsilon}{1+\varepsilon}mg$$

The tension vanishes when there is no moment of inertia $\varepsilon = 0$ *and is maximal for the hoop* $T_{hoop} = \frac{1}{2}mg$.

Note that many of the formulae in the example above hold also for other geometries. We need only change the value of ε.

Example 22. *Consider again figure 22 and suppose that it shows a beam of mass* $M = 30$ *kg and length* $\ell = 3.0$ *m balancing two masses* $m_1 = 20$ *kg and* $m_2 = 50$ *kg. At what value of* d *must the fulcrum be?*

Solution. *There are multiple ways to solve this problem. We solve it two ways first using the center of mass and second using torque.*

There are three masses at the following three distances:

- *the rod has mass M at the center of mass* $\frac{\ell}{2}$ *from the right end*

- *mass m_1 is at distance ℓ from the right end*
- *mass m_2 is at distance 0 from the right end*

If we put the fulcrum at the center of mass of this whole system, then all the torques will cancel. This is because gravity acts on the center of mass of the whole system and if the fulcrum is there, then this force cannot generate a torque. (There is no torque for a force through the pivot since the moment arm $r_\perp = 0$ for such forces.)

Let's take the origin to be at m_2 and the positive direction to be to the left. (We are choosing this because the diagram looks like that would be a good choice.) Then the center of mass is a distance

$$x_{CM} = \frac{1}{m_1 + m_2 + M}\left[m_1 L + M\frac{L}{2} + m_2 0\right] = \frac{m_1 + \frac{M}{2}}{m_1 + m_2 + M}L$$

from the right end of the rod. Thus

$$d = \frac{20 + 15}{100}L = 0.35L = 0.35 \times 3.0\ m = 1.1\ m$$

Note that this is less than the distance to the middle of the rod, as it should be.

Let's check this explicitly with torque. The fulcrum is at a distance d from the right end of the rod. Taking counter-clockwise to be the positive direction (check with the RHR that this corresponds to the direction out of the page), we have the following three torques:

- *The mass m_2 generates a negative torque of size $\tau_2 = \mathrm{d}m_2\mathrm{g}$.*
- *The mass m_1 generates a positive torque of size $\tau_1 = (L - \mathrm{d})m_1\mathrm{g}$, since it is a distance $\ell - \mathrm{d}$ from the fulcrum on the other side.*
- *The mass M generates a positive torque of size $\tau_M = \left(\frac{\ell}{2} - \mathrm{d}\right)M\mathrm{g}$, since the center of mass of the rod is a distance $\frac{\ell}{2} - \mathrm{d}$.*

There are no sine factors because the weight vectors are all perpendicular to the displacement vectors. These should all add up to $\tau_{net} = 0$ since the rod is balanced (no angular acceleration).

$$(\ell - \mathrm{d})m_1\mathrm{g} + \left(\frac{\ell}{2} - \mathrm{d}\right)M\mathrm{g} - \mathrm{d}m_2\mathrm{g} = 0$$

At this point we can plug in numbers, but we push through with the symbolic derivation instead. Rearranging and dropping an overall factor of g gives.

$$\left(m_1 + \frac{M}{2}\right)\ell = (m_1 + M + m_2)\,\mathrm{d}$$

so we again recover our formula above for the center of mass.

Note that in the last example, we learned the following

Principle. *To balance an arrangement of masses, it suffices to find the center of mass and place the fulcrum there.*

Example 23. *A frictionless pulley with rotational inertia I and radius r is attached to the ceiling and a mass m hangs from it a height h above the floor, as in figure 28. If the mass is released from rest*

1. *What is the velocity of the mass just before it reaches the floor?*
2. *What is the time it takes to reach the floor?*
3. *What is the tension in the cord?*

Solution. *For the first, we can use conservation of energy, and for the second we can use CAKE, but we already know we will have to use torque for the third. We'll do both.*

The work done by gravity is mgh. By the work-KE theorem this is converted into the final kinetic energy $K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = \frac{m}{2}\left(1 + \frac{I}{mr^2}\right)v^2$ where (as always!) we used rolling without slipping ($v = r\omega$). Equating and solving for v gives (check this!)

$$v = \sqrt{\frac{2gh}{1 + \frac{I}{mr^2}}}$$

This looks like the usual result, but we must be mindful that the hanging mass m and the rotational inertia are unrelated. In particular $\frac{I}{mr^2}$ is not *some geometric factor ε here!*

For the second part, we can use CAKE. Those all have linear accelerations in them, so first we need to find that from CAKE3: $v^2 = 0 + 2ah$ gives $a = \frac{v^2}{2h}$. This goes in CAKE1 (or 2) $v = 0 + a\Delta t$ or $\Delta t = \frac{v}{a} = v\frac{2h}{v^2} = \frac{2h}{v}$. This is (at least) the second time we have derived this useful formula, so let's write it again

$$\Delta t = \frac{2h}{v} \tag{18}$$

Plugging in our result for the speed of the block (check!),

$$\Delta t = \sqrt{1 + \frac{I}{mr^2}}\sqrt{\frac{2h}{g}}$$

Again this looks familiar (as it should!) from (16) and (17), but we must keep in mind that the variables in each situation have different meanings.

To get the tension, we do a torque analysis. Like all force analyses, this starts with a FBD. (You should draw this diagram!) For the mass, the procedure gives Newton II in the form

$$ma = mg - T$$

The torque the mass exerts on the pulley gives the rotational analog (check!)

$$I\alpha = rT \qquad \Rightarrow \qquad Ia = r^2 T$$

We can check from this that

$$a = \frac{1}{1 + \frac{I}{mr^2}} g$$

but we are really after the torque instead:

$$mg = \left(1 + \frac{mr^2}{I}\right) T \qquad \Rightarrow \qquad T = \frac{mg}{1 + \frac{mr^2}{I}} = \frac{I}{I + mr^2} mg$$

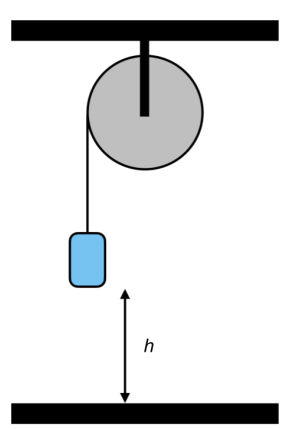
This has the correct units and the properties that $T \to 0$ if either $m \to 0$ or $I \to 0$.

Example 24. *Consider again example 22, but this time, let us assume that the system in figure 22 is not balanced. In fact, let us take $d = 2.0\ m$. Calculate the angular acceleration that results if the system is released from its horizontal position from rest.*

Solution. *We must calculate the torque around the fulcrum, which is to the left of the center of mass now (despite what it looks like in the figure). We expect the system to rotate clockwise (angular acceleration into the page), so let us refer to that direction as positive. Then the torque on the system is*

$$\tau_{net} = -m_1 g(\ell - d) + Mg\left(d - \frac{\ell}{2}\right) + m_2 g d$$

All three terms are the weights multiplied by the moment arms. If we plug in the numbers, we find $\tau_{net} = 3380\ N \cdot m$. Newton's Second Law of motion for rotations states that $\boldsymbol{\tau}_{net} = I\boldsymbol{\alpha}$. Consulting example 14, we can obtain the rotational inertia as $I = \frac{1}{12}M\ell^2 + M\left(\frac{\ell}{2} - d\right)^2 + m_1(\ell - d)^2 + m_2 d^2 = 250\ kg \cdot m^2$. With this, we find an angular acceleration $\alpha = \frac{\tau_{net}}{I} = 13.5\ \frac{rad}{s^2}$.

5.6 Synthesis

We have seen that